

GIM v20.1.0: a closed economy–climate–society world model with measured boundedness at country resolution

Nazar S. Sotiriadi^{1*}, Alexander N. Krenke², Andrei A. Osipov^{1,3}

¹JSC Sberbank of Russia, Kutuzovsky avenue 32, 121170 Moscow, Russian Federation

²Institute of Geography, Russian Academy of Sciences, Staromonetny lane 29 bld. 4, 119017 Moscow, Russian Federation

³Skolkovo Institute of Science and Technology, Bolshoy Boulevard 30 bld. 1, 121205 Moscow, Russian Federation

Abstract

Coupling the economy, climate, society and geopolitics in one closed simulator is easy to attempt and hard to sustain: such loops generically diverge, stick in absorbing states, or must be damped into triviality. We describe GIM v20.1.0, a global integrated model of 57 country agents and seven coupled domains advanced by a deterministic annual map, and we report what its closed loop actually does rather than asserting that it is stable. Numerical linearization of the annual map on a 408-dimensional projection of the state gives a spectral radius of 1.036–1.100 over 31–36 above-unity modes whose eigenvector mass sits on economic coordinates (public debt, unemployment, inflation); the climate coordinates carry none of it, although the infinite-lifetime carbon pool places a unit eigenvalue there by construction. Twin runs contract at 0.975 and 0.970 per year for output and temperature perturbations, while a broad social-tension perturbation amplifies to a 2.5 % world-product gap over thirty years. A 500-member prior ensemble grows near-linearly to $\pm 10.8\%$ of median world product at thirty years. Retrospective evaluation against observations gives in-sample RMSEs of 0.60 trillion USD (country-level output), 0.94 Gt CO₂ and 0.099 °C; on a 2020–2023 window held out procedurally from the calibration filter the model scores 0.77, 0.72 and 0.13, beating persistence and linear extrapolation on emissions and temperature and losing to both on output. Three resource prices beat a flat-price null. We report two negative results in full: the history-matching filter retains 100 % of prior draws, so the ensemble is a prior ensemble with a non-binding diagnostic rather than a history-matched one; and the fixed-weight conflict composite (AUC 0.739) does not beat its own strongest input (AUC 0.776, paired difference -0.037). A single-player decision game is used as an adversarial sampler of the tails and found two structural degeneracies that no retrospective test could see. GIM is a bounded, anchored, responsive exploration instrument, not a forecast engine, and is released under Apache-2.0 with the harness behind every number in this paper.

Keywords: integrated assessment model; coupled human–Earth system; model stability; uncertainty quantification; history matching; adversarial stress testing.

*Correspondence: sotiriadi.n.s@gmail.com

1 Introduction

Strategic planning and systemic-risk work need a tool in which climatic, economic and socio-political processes are considered jointly: the paths that matter — warming to crop yields to prices to social tension to instability — cross domain boundaries that sectoral models do not (Nordhaus, 2017; Riahi et al., 2017). That gap is well known. What is less discussed is why it persists. Coupling the domains is not conceptually difficult; the difficulty is that a *closed* loop through economy, climate, society and politics is a high-dimensional nonlinear dynamical system, and such systems do not stay plausible by default (May, 1972; Sterman, 2000). In our experience they fail in a few characteristic ways: variables run to a bound and stick, accounting errors compound into runaway growth, latent variables drift to attractors no policy can move, or — the opposite failure — every feedback is damped until the simulator reproduces a trend line and nothing else. A world model earns its keep between these failures: bounded for decades, yet still able to generate crises, cascades and lags.

This paper is a model description of GIM v20.1.0 and a report of what the closed loop does. GIM advances 57 country agents (the 50 largest economies plus seven regional aggregates that conserve world totals) through an annual deterministic map spanning the economy and prices, climate and the carbon cycle, natural resources, society and politics, interstate relations, and discrete crises with stock-flow-consistent finance. Its headline property is not the breadth of coverage but three measured behaviours:

boundedness — thirty-year trajectories stay bounded, physical and economic perturbations contract, and ensemble spread grows near-linearly rather than exponentially, while a socio-fiscal subspace amplifies slowly (Section 4); *anchoring* — the same configuration reproduces 2015–2023 in-sample, is scored free-running on a procedurally held-out 2020–2023 window, and carries out-of-sample skill where structure matters (Section 5); and *responsiveness* — lagged social channels, threshold-stepped sovereign cascades and cross-block synchrony of failures whose shapes go beyond the coded terms they build on (Section 6).

We are deliberate about what those words are licensed to mean. We do *not* claim the annual map is locally asymptotically stable: it is not, and its own spectrum says so. What the evidence supports is boundedness over the tested horizon with contracting physical-economic perturbations and slowly amplifying socio-fiscal modes — weaker, and more useful.

The stance behind the first is methodological: the deterministic core is an explicit hypothesis about physical conservation laws, accounting identities and slow empirical regularities (Section 2), so determinism is a feature rather than a naivety — the core computes what follows *if* those relations hold, and parametric uncertainty is carried by ensembles over their strengths. On top sits an instrument layer: prior ensembles, weak-signal detectors, and a decision game used as an adversarial sampler of the tails (Section 7). The model does not claim predictive foresight; beyond two to three years professional growth forecasts lose to a naive mean-growth benchmark in roughly half of countries (Celasun et al., 2021), and we hold this model to the same bar.

Relative to the integrated-assessment and world-modelling traditions — DICE/RICE (Nordhaus, 2017; Nordhaus and Yang, 1996) and FUND (Anthoff and Tol, 2013) (welfare optimisation), GCAM (Calvin et al., 2019) and MESSAGEix (Huppmann et al., 2019) (energy-system equilibrium), E3ME (Mercure et al., 2018) (econometric simulation), T21/iSDG (Millennium Institute, 2002) (single-country system dynamics), World3 (Meadows et al., 1972) (global aggregates) — GIM’s distinction is that it closes the loop through society, geopolitics and discrete crises at country resolution and then *measures* what the

closed loop does. A controlled benchmark characterising the cross-sector transfer functions is in Appendix B; the full specification is in Appendix A. Our evaluation follows the hierarchy set out for environmental and integrated-assessment models (Jakeman et al., 2006; Bennett et al., 2013; Schwanitz, 2013; Wilson et al., 2021), and we adopt the “evaluation” distinction between verification of the implementation and corroboration against data (Augusiak et al., 2014; Grimm et al., 2014).

Of the six persistent critiques of IAMs identified by Keppo et al. (2021), GIM addresses four directly — heterogeneous actors, technology diffusion, capital markets and energy–economy feedbacks — through country agents with bilateral relations, an endogenous R&D stock with trade-driven diffusion, stock-flow-consistent finance with sovereign crises, and a single loop coupling energy prices, inflation and social tension. The remaining two, policy differentiation and result interpretation, are taken up in Sections 6 and 7.

2 Constraints, identities and restoring forces

Every long-horizon simulator embodies assumptions; ours are organised deliberately into three classes in descending order of confidence, plus a fourth element that keeps the loop near them. We previously called all four “invariants”; that word overstates the epistemic status of the third class and we no longer use it as a blanket term.

Physical conservation. Emissions are mass-partitioned: each year’s emissions are distributed over the four IPCC AR6 impulse-response pools and decay on fixed timescales; atmospheric stock, radiative forcing and the two-reservoir surface–ocean heat balance follow standard reduced-complexity climate physics (Joos et al., 2013; Geoffroy et al., 2013; IPCC, 2021; Forster et al., 2021), the same architecture as the FaIR and MAGICC emulators (Smith et al., 2018; Meinshausen et al., 2011), with the emission record anchored to the Global Carbon Budget (Friedlingstein et al., 2023). This is *partition*, not a closed carbon budget: removal is to implicit land–ocean sinks whose stocks the model does not carry, so the identity $E = \Delta C_{\text{atm}} + \Delta C_{\text{ocean}} + \Delta C_{\text{land}}$ cannot be checked inside GIM. What is checked at runtime is that the atmospheric pools receive exactly the emitted mass, $\sum_i a_i = 1$. The block reproduces both constants an AR6-consistent emulator must match: equilibrium sensitivity 3.0°C and transient response 1.79°C (AR6 best estimate 1.8).

Accounting identities. Money and goods cannot appear from nowhere. In GIM’s simplified banking balance sheet every loan creates a matching deposit, so within the model the money stock is identically the sum of deposits and loans; this is a model accounting identity in the stock-flow-consistent tradition (Godley and Lavoie, 2007), not a claim about the monetary aggregates of any real economy. Government budgets close; bilateral trade settles in reserves and world net exports sum to zero by an enforced constraint (residual 4×10^{-18} of world product); excessive borrowing raises the cost of credit — a financial accelerator (Bernanke et al., 1999). These identities are checked at runtime, not assumed.

Empirical regularities as slow variables. Between hard conservation and free parameters sits a class of relationships that are not laws but have been stable across decades and countries; we treat them as constraints with priors on their strength. Trade follows a gravity law in distance (Head and Mayer, 2014); unemployment follows Okun’s law (Okun, 1962) and inflation a flat post-1990 Phillips curve (Phillips, 1958) with a weak quantity-theory term; poorer economies converge toward the frontier at a β -convergence rate validated on

the cross-section; the public’s response to unemployment and inflation follows the two-to-one misery-index ratio (Di Tella et al., 2001); war sizes follow a power law (Richardson, 1948; Clauset et al., 2009); institutions are anchored to governance indices and culture to the Hofstede dimensions (Kaufmann et al., 2011; Hofstede et al., 2010); monetary policy follows country-level Taylor rules (Taylor, 1993). Where a regularity failed the data — a Richardson arms-race term, rejected at $p = 0.74$ on the 1995–2024 military-expenditure panel — it was left out and the observed state variable used instead.

Equilibrium anchors and restoring forces. Constraints alone do not prevent drift; bounded behaviour comes from explicit restoring terms whose strengths are stated rather than hidden: resource prices mean-revert in log space toward reference levels; production is normalised so each country’s base point is exactly reproduced (a normalised-CES discipline that pins levels while leaving substitution free); trust and tension carry anchor terms; crisis states have exit conditions. Two of these restoring terms are behavioural regularisation rather than empirical calibration, and Table 2 labels them as such. Section 4 shows what these forces achieve and Section 7 how two missing anchors were found the hard way.

Determinism, in this view, is the point: the core is a reproducible mapping from a stated set of relations to trajectories. Following Oreskes et al. (1994), we do not claim the model is verified truth — we claim it is a confirmed, auditable encoding of stated assumptions, evaluated by whether its trajectories stay within the envelope of observed history and whether its failures are informative (Saltelli et al., 2020).

3 Model description

3.1 Agents, domains and the annual loop

One model step is one year; blocks update in a fixed order and the year’s outputs are the next year’s inputs (Fig. 1). The economy is the hub: output comes from capital, labour and energy through a nested CES function (Eq. A1), and inflation, unemployment, budgets, debt and bilateral trade are tracked per country. Output emits CO₂ at a declining structural intensity; emissions accumulate in the carbon pools; warming returns as a quadratic output damage. Resource stocks and world prices couple scarcity into inflation. Society tracks trust, tension, inequality, legitimacy and protest pressure; politics converts legitimacy into a policy-space multiplier; geopolitics carries relations, alliances, sanctions and a CINC-style capability index (Singer et al., 1972) grounded in SIPRI expenditure (SIPRI, 2024). Discrete crises — debt, currency, regime collapse — fire on threshold crossings, apply audited state changes and have exit rules, and conflict, unrest and climate risk spill over to geographic neighbours. Functional detail is deferred to Appendix A; what matters here is that the loop mixes amplifying channels (financial accelerator, crisis contagion, tension–trust spiral) with the restoring forces of Section 2, and the balance between them is what Section 4 measures.

The 2023 base-year world product of the 57 agents is 106.8 trillion USD. Units are stated once and used throughout: country output levels are anchored to 2015 current US dollars at market exchange rates and carried forward by *real* growth taken from the World Bank constant-price (2021 international \$, PPP) growth index, so all output figures in this paper are real quantities on a 2015 market-exchange-rate base, not nominal and not PPP levels. On that basis the thirty-year deterministic baseline reaches 263.8 trillion USD in 2053, a compound real growth rate of 3.06 %/yr.

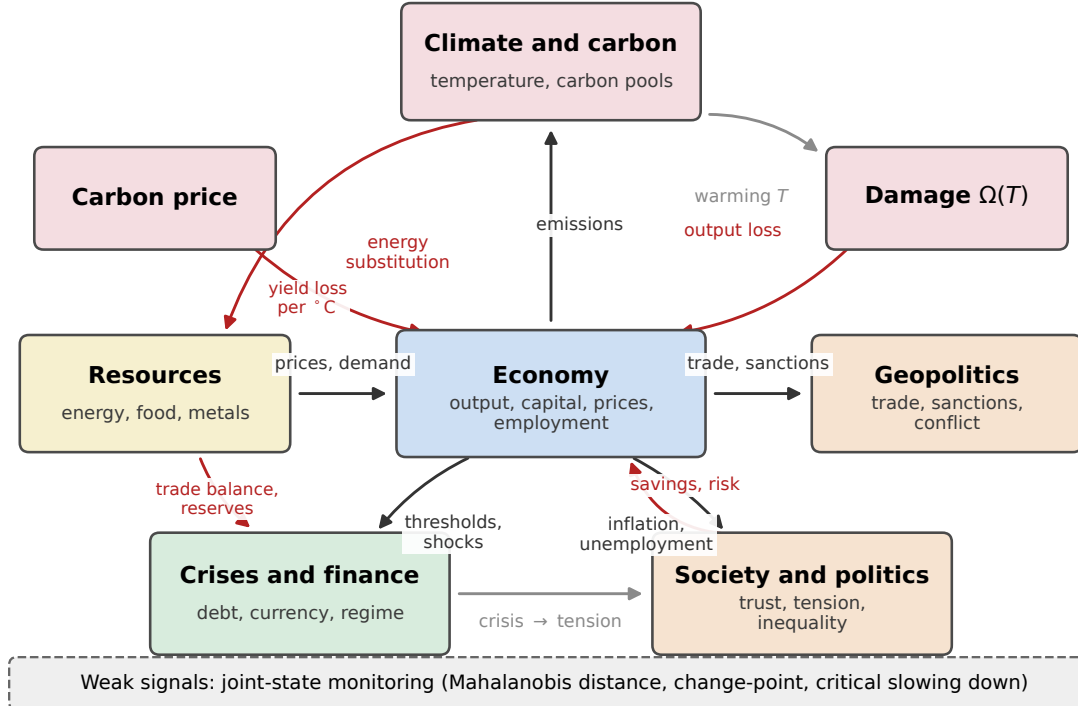


Figure 1: The closed loop. Economy at the hub, physical blocks above it and human blocks below. Dark arrows are direct effects; red arrows are the channels that reduce output, savings or supply — climate damage, energy substitution, the per-degree yield loss on crops, and the resource trade balance that drives reserves and the sovereign channel. The climate loop (emissions $\rightarrow$ warming $\rightarrow$ damage $\rightarrow$ output) is complemented by carbon-price, resource, socio-political, geopolitical and discrete-crisis channels. The bottom layer monitors the joint state for weak signals.

3.2 Numerical implementation and the annual algorithm

In a threshold-carrying nonlinear model the update *order* is part of the numerical specification, not an implementation detail, so we give it in full. Algorithm 1 is the effective write order of one call to `step_world()` in `gim/core/simulation.py`; the same listing is maintained as `docs/SIMULATION_STEP_ORDER.md` in the archive and is checked by a test.

Table 1: Algorithm 1: one annual GIM step, $x_t \mapsto x_{t+1}$. Steps run once, in this order, with no intra-year iteration and no market-clearing fixed point; the four-phase scaffold on the right is the runtime’s write-authority contract.

#	Operation	Phase
1	refresh relative metrics (shares, ratios, gaps)	baseline
2	update political states; update institutions	baseline
3	generate policies; apply political constraints	baseline
4	resolve foreign policy	detect
5	apply sanctions effects; apply security actions	propagate
6	update active conflicts; apply trade-barrier effects	propagate
7	apply policy actions; apply trade deals; update endogenous relations	propagate
8	allocate energy reserves and caps; update resource stocks	propagate
9	update global resource prices	propagate

Table 1 — *continued*

#	Operation	Phase
10	update global climate (carbon pools, forcing, two-layer heat balance)	propagate
11	update climate risks; apply climate extreme events (off in headline runs)	propagate
12	update economy output (CES, damage, capital)	propagate
13	update public finances (budget, debt, credit cost)	propagate
14	check financial crises: debt trigger, then FX trigger	propagate
15	update inflation and unemployment (Phillips, Okun, energy and food cost-push)	propagate
16	update migration flows; update population	propagate
17	update social state (trust, tension, inequality, legitimacy, protest)	propagate
18	check regime stability	propagate
19	canonical finalisation of critical fields; clamp to declared bounds	reconcile
20	refresh relative metrics; update agent memory; update credit ratings	reconcile
21	$t \leftarrow t + 1$	reconcile

Integration and time stepping. Only the climate block carries continuous-time dynamics. The four-pool carbon impulse response is integrated exactly over the annual step ($C_{t+1}^{(i)} = C_t^{(i)} e^{-1/\tau_i} + a_i E_t$, Eq. A6), so no discretisation error enters the carbon stock. The two-layer energy balance (Eq. A7) is advanced by explicit Euler at $\Delta t = 1$ yr. With the headline surface heat capacity $C_s = 8 \text{ W yr m}^{-2} \text{ K}^{-1}$ and feedback $\lambda = 1.24 \text{ W m}^{-2} \text{ K}^{-1}$ the stability limit of that scheme is $\Delta t < 2C_s/(\lambda + \kappa) = 7.1 \text{ yr}$, so the annual step is comfortably inside it across the whole prior box: the tightest corner sampled ($C_s = 6$, ECS = 1.5°C giving $\lambda = 2.47$, $\kappa = 1.3$) still gives a limit of 3.2 yr , and the high-sensitivity corner the reviewer of such a model will ask about ($C_s = 18$, ECS = 6°C) gives 35 yr . Accuracy, not stability, is therefore the binding question, and it is small: re-integrating the baseline forcing path at $\Delta t = 1/12 \text{ yr}$ changes the 2100 surface temperature by 0.002°C at the headline parameters and by at most 0.012°C at any corner of the prior box — an order of magnitude below the in-sample temperature RMSE of Section 5. Every other block is a first-order difference equation in annual time; there is no inner iteration and no convergence criterion to report, because the annual map is evaluated once rather than solved to a fixed point. Within-year market clearing is therefore imposed by construction rather than converged to, a consequence Section 8 carries.

Thresholds and discrete events. Crisis triggers are hard comparisons on state ratios (Appendix A), so the annual map is piecewise smooth with jump discontinuities on the trigger manifolds. This matters for the numerical Jacobian of Section 4: a central-difference stencil that straddles a trigger returns a meaningless derivative. We therefore verify, at each linearization point, that no trigger changes state anywhere in the stencil (it does not, at $\varepsilon = 10^{-4}$ of coordinate scale), and we report the sensitivity of the spectral radius to ε .

Reproducibility and verification. The engine is deterministic given a seed: repeated runs of the same configuration are bit-identical on the same platform, and the regression suite pins golden trajectories that any change to the core must either reproduce to the bit or explicitly re-baseline with a written reason (`scripts/regenerate_backtest_baseline.py` refuses to write when a guarded metric regresses by more than 1% without an override flag). The suite carries 592 tests and 657 subtests, including a ten-year 57-agent golden world run and a guard that every

crisis channel fires at least once; it runs in about two minutes on a consumer laptop. Calibration inputs are checksum-bound and every run writes a manifest recording the code version, the input digests and the random seeds. Following the “evaluation” distinction (Augusiak et al., 2014), we treat this apparatus as *verification* — evidence that the implementation computes the intended equations — and reserve *corroboration* for Section 5. Cross-platform bit-identity is not claimed: results are reproducible to floating-point tolerance across platforms and bit-identical within one.

3.3 Configuration: which mechanisms are active

GIM ships optional mechanisms as switches, off by default unless anchored by evidence. Because several of the paper’s claims depend on which switches are set, Table 2 gives the complete headline configuration and states where each setting is used. Every number in this paper other than those explicitly marked as ablations was produced with this configuration.

Table 2: Headline configuration. “Provenance” distinguishes empirical calibration against observations, structural normalisation, and behavioural or numerical regularisation — three different epistemic statuses that a single word “calibrated” would hide.

Mechanism	Headline	Where it enters	Provenance
Extreme climate events	off	all deterministic results; on for the seeded ensemble of Sec. 6	structural
Continuous climate→tension term (TENSION_CLIMATE_SENS)	off	nowhere; magnitude not identified (Hsiang et al., 2013; Buhaug et al., 2014)	tagged, unset
Trust-gap effect on tension $\sigma_{tr} = 0.060$ (Eq. A10)	on	all runs, incl. the spectra of Fig. 2	empirical (WGI trust persistence)
Trust mean-reversion pull $\chi = 0.10 \text{ yr}^{-1}$ (Eq. A9)	on	all runs, incl. the spectra of Fig. 2	<i>numerical regularisation:</i> weakest value lowering ρ at all three linearization points
Reserve-accumulation identity	off	disabling it is a disclosed limitation (Sec. 8)	structural (no nominal FX rate)
Growth-damage (Burke-type) TFP channel	off	tail scenarios only (Burke et al., 2015, 2023)	tail prior
Warming-benefit cap; CH ₄ feedback	off (0)	tail scenarios only	tail prior
Model-consistent expectations	off	tested, marginal effect unstable	structural
LLM policy agents	off	never in a validated configuration	structural
Spatial diffusion (tension 0.05, climate 0.10, contagion 0.03)	on	Sec. 6 ablations	expert prior
SSP2 productivity anchor	on	long-horizon baseline; ablated in Sec. 5	scenario input

Table 2 — *continued*

Mechanism	Headline	Where it enters	Provenance
Non-CO ₂ forcing	exogenous	SSP2-4.5 marker (Meinshausen et al., 2020)	scenario input

Two entries in that table deserve to be named as what they are. The trust mean-reversion pull was selected as the weakest value that lowers the spectral radius at all three linearization points — that is, it was chosen *on the diagnostic that Section 4 then reports*. This is circular if read as evidence that the loop is stable, and we do not read it that way: the honest statement is that a bounded institutional variable with no restoring force carries a unit root by construction, that GIM supplies the weakest restoring force that removes it, and that the spectra we report are those of the resulting map. Section 4 gives the spectra at four settings of that pull, including zero, so the reader can see exactly how much of the result the choice buys.

4 What the closed loop does: boundedness, measured

Convergence claims are cheap, and we do not make one. This section reports three quantitative diagnostics and states exactly what each licenses. All runs use the deterministic core in the configuration of Table 2; the harness is `scripts/run_stability_analysis.py`.

4.1 Spectrum of the one-year map

We linearize the annual map $x_{t+1} = F(x_t)$ numerically by central differences around the baseline trajectory at three points along the run. The coordinates differentiated are seven variables per agent (output, capital, public debt, unemployment, inflation, trust, tension) plus nine global coordinates (surface and ocean temperature, four carbon pools, three resource prices): $57 \times 7 + 9 = 408$ in all. Eigenvalues are computed on $D^{-1}JD$ with D the diagonal of trajectory-mean coordinate scales — a similarity transform that leaves the spectrum unchanged and only conditions the eigenvector readout.

What this Jacobian is, and is not. It is *not* the Jacobian of the full Markov state. GIM’s persistent state registry carries 87 variable families (`docs/state_registry.csv`), including reserves, bilateral relations, alliances, sanction counters, resource stocks, credit ratings, population and the crisis-duration counters. Perturbing one of the 408 coordinates advances *all* of them for one year, but only the 408 are read back, with the complementary coordinates held at their baseline values. What we compute is therefore the Jacobian of the projected map $P \circ F \circ \iota_{\bar{x}}$ — a principal submatrix of the full Jacobian, restricted to the coordinates that carry the model’s stocks and prices. Its spectrum bounds the behaviour of perturbations confined to that subspace; it does not bound perturbations that excite reserves, relations or crisis counters directly. We state this as a limitation rather than repair it, because differentiating the full registry would require a state serialisation the engine does not currently expose, and we flag it as the first item of future work.

Result. The spectral radius is $\rho(J) = 1.036, 1.100, 1.041$ at 2026, 2036 and 2046, over 36, 35 and 31 above-unity modes (Fig. 2a). Two properties of that spectrum set what a stability claim about a model of this kind can assert.

The first concerns a large cluster of modes at unit modulus: 45, 12 and 24 eigenvalues lie inside $[0.999, 1.001]$ at the three points, and 94 or more lie above 0.99. Part of that cluster is a specification requirement rather than a result, because a bounded state variable with no restoring force of its own carries a unit eigenvalue by construction. Institutional trust is the case in point: bounded in $[0, 1]$, its drivers must enter as deviations from a reference or an agent at entirely ordinary conditions loses trust every year without limit. The deviation form removes that drift but not the unit root, since a walk without drift still sits at $|\lambda| = 1$; GIM therefore gives trust the weak mean reversion its resource prices already carry (Table 2), and we report the spectrum of the resulting map. The first carbon pool is the second and physically correct case: with $\tau_1 = \infty$ it is a perfect integrator, $\partial C_{t+1}^{(1)} / \partial C_t^{(1)} = 1$ exactly, so the climate block *contains a conservative direction by construction*. That is a property of the carbon cycle, not a defect — but it makes “the climate subspace contracts” false.

The second is the result. Modes above unity remain — the loop does not contract everywhere — but their eigenvector mass sits mainly on *economic* coordinates: public debt compounding near the effective interest rate, unemployment and inflation carry 0.57, 0.65 and 0.78 of the above-unity loading at the three linearization points, against the social block’s 0.43, 0.35 and 0.22. The climate coordinates carry *none* of the above-unity mass: across every configuration we examined — four settings of the trust mean-reversion pull at three linearization points, twelve spectra in all — the above-unity mass on temperature, ocean heat and the carbon pools is 0.00 to two decimals. The defensible statement is therefore that *the climate coordinates are non-expanding and carry a conservative direction*, not that they contract, and that the amplifying directions of this model at this calibration are socio-fiscal. We report it as a property of this model at this calibration, not as a general feature of coupled world models.

Sensitivity of the diagnostic. Two robustness checks belong with the number. First, the trust mean-reversion pull, which Table 2 flags as numerical regularisation: at pull = 0 the radius is 1.075, 1.100, 1.076 over 75, 79, 67 above-unity modes; at 0.05, 1.052, 1.100, 1.100 over 50, 48, 49; at the headline 0.10, 1.036, 1.100, 1.041 over 38, 35, 32; at 0.20, 1.020, 1.100, 1.100 over 34, 29, 32. The pull roughly halves the count of above-unity modes and moves the radius by at most 0.04; it does not change the qualitative picture, and the climate mass is 0.00 at every setting. Second, the differentiation step. Because the map is piecewise smooth, a central-difference stencil straddling a crisis trigger would return a meaningless derivative; we verified that no trigger changes state anywhere in the stencil at the headline $\varepsilon = 10^{-4}$ of coordinate scale, and recomputed the whole spectrum at $\varepsilon \in \{10^{-3}, 10^{-4}, 10^{-5}\}$ at the 2026 linearization point. The radius is 1.03631 at all three, identical to five decimal places over two orders of magnitude in the step, and the count of modes above 1.001 is 32 at all three. Only the raw count above 1.000 moves (36, 36, 37), because 45 eigenvalues sit inside $[0.999, 1.001]$ and a handful cross the line under rounding. We therefore quote the radius as a converged result and the above-unity mode count as an approximate descriptor, and recommend the same caution to anyone reporting above-unity counts for a model whose spectrum is this dense at the unit circle.

4.2 Perturbation contraction

The trajectory-level counterpart: twin runs from ε -perturbed initial states, gaps measured as dimensionless relative differences in world product. A +0.1% shock to United States

output — a vector with essentially no projection on the socio-fiscal modes above — produces a relative gap of 2.6×10^{-4} after one year that *contracts* to 1.2×10^{-4} by year 30, an empirical factor of 0.975 per year; a $+0.01^\circ\text{C}$ temperature perturbation decays at 0.970 per year. Nearby trajectories converge in the economic–climate subspace — the opposite of chaotic divergence (Fig. 2b). The one persistent direction is the one the spectrum names: raising every country’s tension by 0.01 leaves output almost untouched for a decade and then amplifies as crisis triggers engage, reaching a 2.5 % world-product gap by 2053.

The spectrum and the twin runs therefore tell one story: contraction in the physical–economic directions, slow amplification confined to the socio-fiscal subspace. We report the asymmetry as a substantive property, not a defect — and it quantifies the limitation that socio-political initial-state error does not wash out (Section 8). Two caveats bound the reading. These are three specific perturbation directions, not a norm bound: for a non-normal J a full-state perturbation can be amplified transiently even when every direction we tested contracts, and we have not computed the finite-time amplification operator $G(t) = \|\delta x(t)\|/\|\delta x(0)\|$ over random full-state perturbations. And the gap metric is the relative world-product gap, which is insensitive by construction to directions that redistribute output without changing its total.

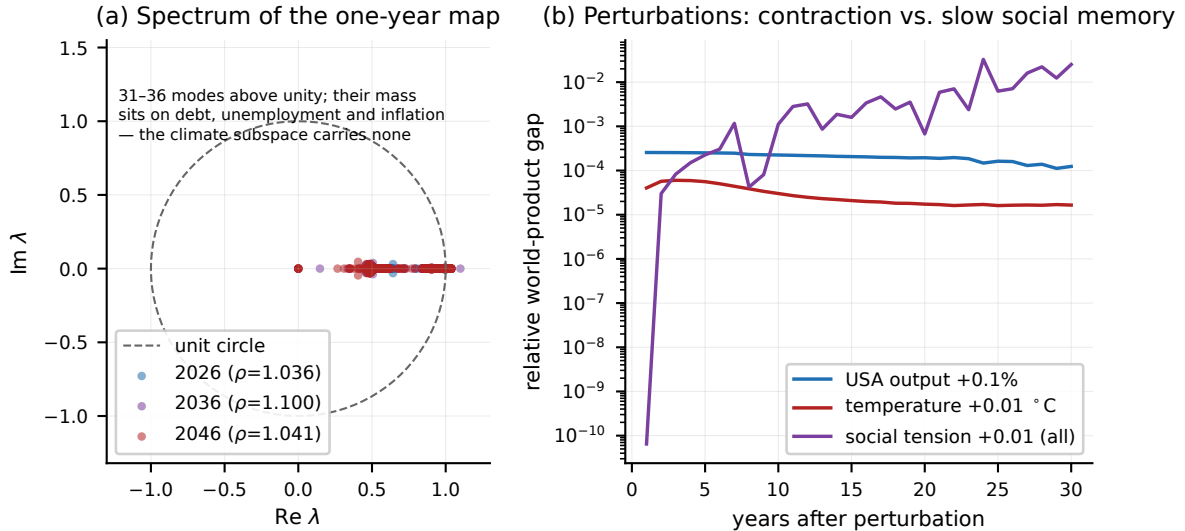


Figure 2: Local behaviour of the annual map. (a) Eigenvalues of the (similarity-) normalized Jacobian of the 408-dimensional projected state at three linearization years. The spectrum lies on and inside the unit circle apart from 31–36 modes just outside it, giving $\rho(J) = 1.036, 1.100, 1.041$; their eigenvector mass sits mainly on economic coordinates — public debt, unemployment, inflation — carrying 0.57–0.78 against the social block’s 0.22–0.43, while the climate coordinates carry none of it. Note the dense cluster at unit modulus: the $\tau = \infty$ carbon pool is a perfect integrator and contributes $|\lambda| = 1$ by construction. (b) Twin-run response (relative world-product gap): output and temperature perturbations contract at 0.975 and 0.970 per year, while a broad social-tension perturbation persists and amplifies — the model’s one long-memory direction.

4.3 Bounded, near-linear growth of parametric spread

A 500-member ensemble drawing the 33 key parameters from their priors — with the discrete-crisis machinery *active* (members average one to three sovereign crisis episodes by 2053) — runs the full 30 years without a member diverging or saturating (Fig. 3).

This is a *parametric, conditional* fan, not a forecast interval. Members share the initial state, so much of its near-linear growth is each member following its own prior-drawn trend, and the fan varies $\partial x_t / \partial \theta$ where the twin runs vary $\partial x_t / \partial x_0$ — different objects. A parametric fan that does not explode is *not* independent evidence of dynamical stability: unstable directions may be poorly excited by parameter perturbations, or invisible in world product. We report it as a bounded parametric sensitivity and nothing more.

What the fan does bound is the amplification rate: the world-product half-width grows from $\pm 1.7\%$ of median at year 5 to $\pm 4.0\%$ at 10 and $\pm 10.8\%$ at 30 (all 5–95th half-widths), an average exponential rate below 0.08 yr^{-1} with concave shape after year 10. Two non-trending observables make the point without the trend caveat: the mean-tension fan stays narrow and sub-linear (5–95th width 0.011 at year 5, 0.026 at 30, i.e. 2–5 % of its level), and the temperature fan widens only from 0.72 to 1.10 °C between years 5 and 30 (full widths; the climate-sensitivity prior expresses itself early). Median 2053 world product is 264 trillion USD (5–95th: 235–292) at +1.96 °C (1.41–2.51); for calibration of scale, a single structural choice — renormalising production to constant returns — moves 2100 world product by 7–9 %, of the same order as this whole parametric fan.

The ensemble is a prior ensemble, and we say so. We previously described this fan as history-matched. It is not, and the diagnostic that would make it so does not bind. The published configuration runs a single wave of 60 Latin-style draws over the six backtest-driving parameters; at the three-sigma implausibility cut the acceptance rate is 100 %, so the not-ruled-out-yet region coincides with the prior box and the ensemble draws all 33 parameters independently from their priors. Over 600 draws from the box the maximum implausibility anywhere is 1.69 against a cut at 3.0, with a median of 0.665; the per-output tolerances are loose by a consistent factor of about five, and world output spans only $I = 0.594\text{--}0.619$ across the entire box, meaning the window does not identify the economic parameters at all. The correct description is a *literature-informed, empirically calibrated and expert-specified prior ensemble with a non-binding history-matching diagnostic*, and Section 5 shows the diagnostic still earns its keep as a *ranking*. Tighter tolerances and multi-wave refocusing are future work; the current tolerances are wide because model discrepancy dominates observational error, and we prefer to publish that than to tighten tolerances until the filter appears to bind.

What “bounded” is and is not. Collecting the three diagnostics: over the tested thirty-year horizon and the configurations examined, GIM’s trajectories are bounded, its physical and economic perturbations contract, and its parametric spread grows near-linearly, while a socio-fiscal subspace amplifies slowly and a conservative carbon direction neither grows nor decays. That is the claim. It is not local asymptotic stability, it is not a statement about the full Markov state, and it is not a proof — the evidence is local, numerical, and specific to this calibration.

5 Evaluation against observations

A bounded simulator can still be stably wrong; the second requirement is anchor to observation. All retrospective runs are *free-running*: initialized once from the 2015 state and never re-anchored to observations, so errors are multi-year trajectory errors, not one-step-ahead errors. The output metric is the RMSE over country-year pairs (20 largest economies) in trillion USD; emissions and temperature are world-series RMSEs against

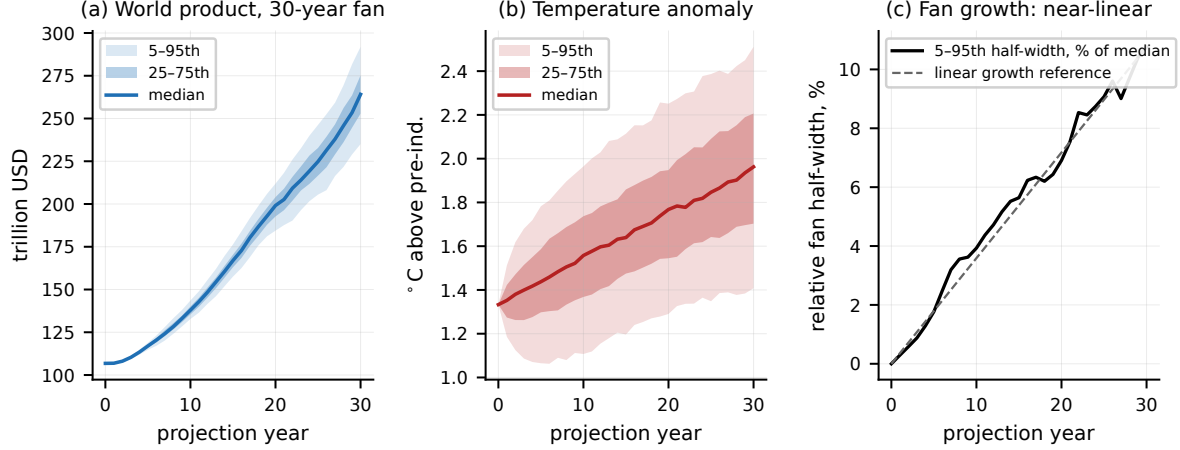


Figure 3: Thirty-year boundedness evidence (500-member prior ensemble, shared initial state, discrete crises active, seed 2026). (a) World product fan: all members bounded, median 264 T at 2053, 5–95th 235–292 T. (b) Temperature fan: full 5–95th width grows $0.72 \rightarrow 1.10$ °C between years 5 and 30, median 1.96 °C at 2053. (c) Relative 5–95th half-width of world product reaches 10.8 % of the median at thirty years and tracks a linear reference — parametric spread accumulates, with no super-exponential bending.

the Global Carbon Budget (Friedlingstein et al., 2023) and HadCRUT5.1.0.0 rebased to 1850–1900 (Morice et al., 2021). Temperature errors are reported for the deterministic configuration (internal interannual variability off), which is the configuration every other result in this paper uses; enabling the variability term raises the in-sample figure from 0.099 to 0.145 °C because the model’s realization is not synchronized with the observed one, and we give both rather than choosing silently.

The economic and climate blocks are calibrated by history matching — ruling out parameter regions whose implausibility $I_o(\theta) = \text{RMSE}_o(\theta)/\text{tol}_o$ exceeds a three-sigma cut, a standard approach for expensive simulators (Craig et al., 1997; Williamson et al., 2013) — on the 2015–2023 world series. Predictive adequacy is then measured by a free-running temporal test on 2020–2023 (`scripts/run_holdout_validation.py`).

How the held-out window is constructed, and what leaks anyway. The test is procedurally clean: history matching is re-run with the implausibility computed *only* on 2015–2019, the retained region is frozen, and 2020–2023 is then scored without re-anchoring. Because the 2015–2019 window rules out no draw at all, the frozen region coincides with the prior box, and the exercise reduces to running the frozen configuration forward. That is a real out-of-sample test of the *parameters*. It is not a clean test of the *model*, and we do not claim it is. Three named channels leak: the structural decarbonisation rate, the energy elasticity and the TFP sensitivity to R&D were all set with reference to the 2015–2023 backtest (Table A1); the resource-price subsystem was repaired after comparison with observed 2015–2023 commodity indices; and functional-form choices throughout the development history were made against the same period. The honest name for what we report is therefore a *free-running retrospective validation with a procedurally held-out scoring window*, not an untouched out-of-sample test. Constructing the latter would require freezing model structure as of 2019 and is not recoverable after the fact.

That the cut does not bind is itself a result. History matching here is a *ranking*, not a filter — and as a ranking it works: per-output implausibility predicts its own out-of-sample error at $\rho = 0.90\text{--}0.95$. The usual $\max_o I_o$ aggregation destroys that information for output ($\rho = +0.06$) because the maximum is almost always taken by temperature or CO₂, whose skill is unrelated to output skill. We report per-output implausibilities separately for that reason, and suggest others do the same for multi-domain simulators.

Results. Table 3 and Fig. 4: in-sample RMSE 0.60 trillion USD (country-level output), 0.94 Gt CO₂, 0.099 °C; on the held-out 2020–2023 window 0.77, 0.72 and 0.13 (with 2020 excluded: 0.83, 0.12, 0.15). The per-year decomposition is more informative than the window averages: output errors are 0.56, 0.78, 0.80, 0.90 across 2020–2023 — a monotone free-running drift, not a pandemic spike, which is why excluding 2020 *raises* the output figure — while emissions show the opposite anatomy, a single COVID-sized miss of 1.42 Gt in 2020 collapsing to 0.05–0.12 Gt in 2021–2023, exactly what a model without a pandemic submodel should produce when output is nearly right and lockdown-specific emission suppression is outside its physics; per-year temperature errors are 0.02–0.17 °C. Skill against naive benchmarks built on pre-2020 data divides by output: temperature beats persistence (+0.05) and linear extrapolation (+0.16), emissions beat both decisively (+0.32, +0.54), output loses to both (−0.42, −3.56) — short smooth aggregates remain the domain of a ruler, consistent with the forecast-evaluation literature (Celasun et al., 2021). The climate block additionally tracks the 1990–2023 record (0.096 °C) with best-fit sensitivity 3.0 °C, consistent with the multi-line assessment of Sherwood et al. (2020).

Resource prices are now a validation target, and were not before. The observed fixture carried output, emissions and temperature only, so the price subsystem sat outside every check in the repository. It now carries the World Bank commodity indices (World Bank, 2024a) renormalized to the base year, with each tolerance set to the error a no-skill *flat* price makes, so the implausibility reads directly: below one beats a flat line. The model scores energy 0.99, food 0.80, metals 0.96. Adding the check exposed three previously invisible defects — recycled secondary supply added on top of a base-year figure that already represented total supply, injecting a permanent metals glut; a price-damping buffer reading the geological reserve rather than tradeable inventory, which for energy is fifty years of production in the ground and froze that price at its initial value for the whole window; and energy demand growing with neither population nor income. The energy result deserves plain reporting: the model now moves that price in the right direction but captures only a fraction of the observed amplitude, because the 2015–2023 series is dominated by the 2020 demand collapse and the 2022 supply shock, neither of which the model represents. A near-flat energy price is correct for a model without those shocks; what was wrong before was that it was flat *by construction* rather than by result.

At country level — where the crisis triggers live — the 2015–2023 per-country output RMSE has median 0.10 and interquartile range 0.05–0.13 trillion USD across the 20 anchored economies, the largest five errors concentrated where levels are largest (China 2.4, United States 1.1, India 0.27, Japan 0.24, United Kingdom 0.17). Crisis triggers compare debt and reserves *ratios* within a country, which damps but does not remove the propagation of level errors into trigger distances — a caveat carried by the scenario results.

5.1 Socio-political blocks, including a negative result

The socio-political blocks are held to a softer but real bar. The conflict-proneness score — a fixed-weight composite of observed inputs with no coefficient estimated from conflict data — ranks the 57 agents against the UCDP/PRIO Armed Conflict Dataset v24.1 for 1990–2023 (Gleditsch et al., 2002; Davies et al., 2024) at AUC 0.739 (95 % CI [0.60, 0.86], 20,000 bootstrap resamples (Efron and Tibshirani, 1993); permutation $p \approx 0.001$); observed conflict incidence falls monotonically across score quintiles: 6/11, 6/11, 5/11, 3/11, 1/13 countries (55 % at the top to 8 % at the bottom — counts this small support ranking, not calibration, claims).

Chance is the wrong benchmark for a fixed-weight composite, and against the right one the composite does not survive: its single dominant input, regime stability (construction weight 0.39), scores AUC 0.776 on its own, and the paired bootstrap difference is -0.037 with a 95 % interval of $[-0.114, +0.031]$ and $P(\text{composite better}) = 0.15$. Social tension alone also edges it (0.749). The military term is *below* chance (0.472) while carrying a 0.20 construction weight. The blend adds nothing detectable over one published governance indicator, and a reader wanting a conflict ranking should use regime stability directly.

We keep the composite in the operational model for a reason worth stating rather than implying: the static test scores it at base-year values, whereas the model consumes it as a *dynamic* function of endogenous state — regime stability, trust, inequality, water stress and climate risk all move during a run, and the composite is how they reach conflict risk. Substituting the best static covariate would sever four of those five channels. That is a design argument, not evidence, and it does not license reporting the composite’s AUC as a success: the negative result is the headline. The Brier skill score against a constant base-rate forecast is $+0.123$, but the composite is a rank-rescaled index rather than a calibrated probability, so that figure is a relative statistic between two equally uncalibrated forecasts.

Because regime stability and military burden are themselves affected by conflict, the score is recomputed in a vintage-clean configuration (inputs from pre-2000 data only, evaluated against 2000–2023 conflicts): AUC 0.84 [0.72, 0.94], evaluated on the 2000–2023 window and hence not directly comparable to the 1990–2023 headline, and identical to the 2023-vintage score on the same subset and window. This shows the ranking is not an artifact of late input vintages. It does *not* establish absence of reverse causation: pre-2000 regime stability may itself reflect earlier conflict or a latent common cause, and the test cannot separate those.

The social cost of carbon cross-checks against the published range: Ramsey central $\approx \$90/\text{tCO}_2$ at 200 years, spanning $\$90\text{--}280$ across (ρ, η) choices, below the EPA-2023 central $\approx \$190$ (U.S. Environmental Protection Agency, 2023; Rennert et al., 2022). Fed DICE’s own damage function the engine returns $\approx \$15$ against DICE’s $\$31$ (Nordhaus, 2018) — confirming the higher native estimate is not produced by an inflated valuation layer; the residual factor of two reflects the 200-year truncation, the different carbon cycle and the different baseline growth path, and we do not claim a full reconciliation.

5.2 The long-horizon baseline, and why we do not present it as a projection

With all headline anchors active, the deterministic baseline reaches 2.75°C in 2100, with the atmospheric stock rising from $\approx 3,270$ to $\approx 4,470 \text{ GtCO}_2$ ($\approx 1,200 \text{ GtCO}_2$ of net airborne accumulation, on a preindustrial reference of $2,184 \text{ GtCO}_2$). Removing the SSP2 productivity anchor — emissions and decarbonisation are endogenous either way — lowers the

no-intervention 2100 outcome to about 2°C. That this is *nearest* the strong-mitigation SSP1-2.6 envelope is a coincidence of level, not scenario consistency: GIM’s non-CO₂ forcing is taken exogenously from the SSP2-4.5 marker (Meinshausen et al., 2020) in both cases, so the comparison is a temperature proximity statement and the construction is a hybrid. The gap between the two is a large, disclosed structural sensitivity: the structural decarbonisation rate fitted to 2015–2023 (1.6%/yr) is optimistic when extrapolated for decades.

The baseline growth path carries the same warning and we state it in numbers. World product compounds at 3.06%/yr to 2053 but at 4.67%/yr over 2053–2100, reaching 2,256 trillion USD in 2100. That acceleration is not a projection we defend; it is what the R&D and convergence machinery produces when extrapolated three times past its calibration window, and it is the clearest reason the recommended use mode is scenario *differences* rather than absolute levels. It also bears on any absolute reading of the game’s climate gate in Section 7.

Table 3: Evaluation summary. Economic and climate blocks: retrospective evaluation, in-sample and free-running 2020–2023 (held-out values after semicolons). Other blocks: cross-checks, benchmark improvement, or channel identification. All values regenerated from release v20.1.0.

Block	Source	Criterion	Result
Economy	PWT (Feenstra et al., 2015), WDI (World Bank, 2024b)	in-sample country-level RMSE 2015–2023; free-running 2020–2023	\$0.60 T; \$0.77 held out
Climate and carbon	AR6 (IPCC, 2021), GCB (Friedlingstein et al., 2023), HadCRUT5 (Morice et al., 2021)	in-sample; free-running 2020–2023; ECS/TCR anchors	CO ₂ 0.94 Gt; 0.72 held out. T 0.099 °C; 0.13 held out
Resources	World Bank commodity indices (World Bank, 2024a)	price RMSE vs a flat-price null (below 1 beats it)	energy 0.99; food 0.80; metals 0.96
Cost of carbon	EPA-2023 (U.S. Environmental Protection Agency, 2023), RFF-SP (Rennert et al., 2022)	published cross-check	\$90 Ramsey central; \$90–280 across (ρ, η)
Conflict	UCDP/PRIO ACD v24.1 (Gleditsch et al., 2002; Davies et al., 2024)	ranking vs. registry; input-vintage test; <i>best single covariate</i>	AUC 0.739 [0.60, 0.86]; vintage-clean 0.84 [0.72, 0.94]; but 1–regime stability alone 0.776, paired $\Delta = -0.037$ [−0.114, +0.031] — the composite adds nothing detectable

Table 3 — continued

Block	Source	Criterion	Result
Social block	SWIID (Solt, 2020), WGI (Kaufmann et al., 2011)	<i>internal</i> channel identification on model output (Sec. 6)	lagged channel identified on the paired impulse response, not on the level (level vs. response $r_s = -0.33$), $n=120$
Boundedness	—	spectrum; twin runs; fan growth (Sec. 4)	$\rho=1.036\text{--}1.100$ over 31–36 above-unity modes, above-unity mass economic (0.57–0.78) with climate mass 0.00; economic and climate directions contract (0.975, 0.970/yr); fan $\pm 10.8\%$ at 30 y

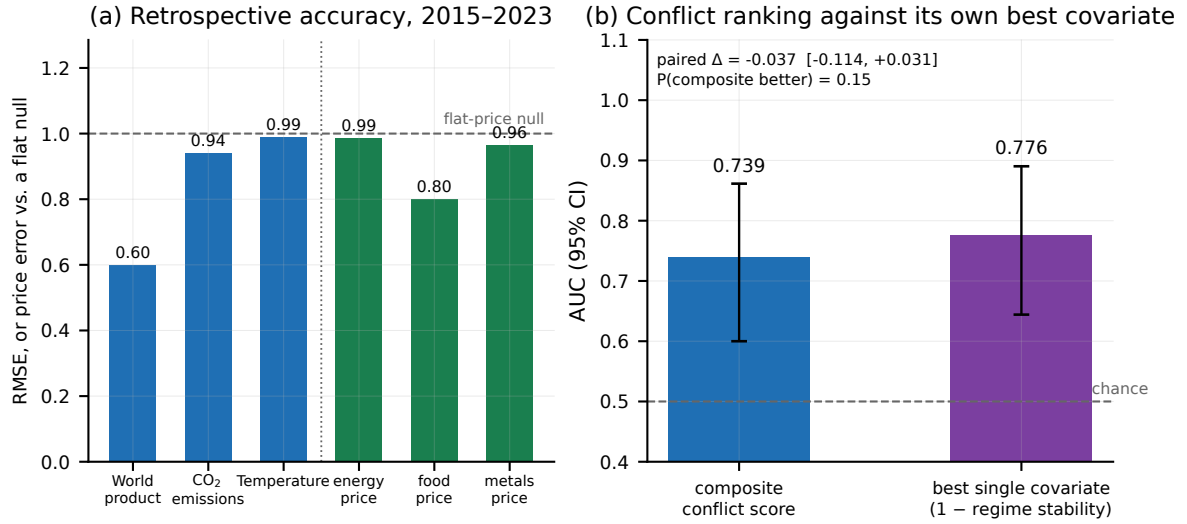


Figure 4: Retrospective anchor. (a) In-sample error 2015–2023 in two different units on one axis, as labelled: world product (trillion USD), emissions (Gt) and temperature (in units of 0.1 °C, so the plotted 0.99 is 0.099 °C) at left; the three resource prices at right, scored as the ratio to the error a no-skill flat price makes, so the dashed line at unity is that null and below it is better. (b) Conflict ranking against the benchmark that decides it — not chance, but the composite’s own best single input. Bars are AUC with 95 % bootstrap intervals; the paired difference and the probability that the composite is the better ranker are inset (score inputs at 2023 base-year values; the vintage-clean variant is given in the text).

6 Cross-domain dynamics

A simulator can be bounded and anchored by being inert. The third requirement is that the loop produce structured behaviour beyond the coded terms it builds on. We separate three things that “emergent” usually conflates: *coded nonlinearity* (a threshold is a threshold), *cross-domain propagation*, and *endogenously generated ordering and timing*. Only the third is genuinely a model output rather than a restatement of the code, and we name in each case which is on offer. Throughout we give effect sizes rather than p -values: on model-generated samples p is a function of the run budget, not evidence about the world. All

causal statements here are *model-internal* — a paired difference $\Delta y = y_{\text{shock}} - y_{\text{baseline}}$ identifies the effect of an intervention inside GIM, which is what a scenario instrument needs and is not a claim about the world.

A lagged economy-to-society channel. Coded ingredient: linear increments to an accumulating tension index, whose multi-year timescale follows from accumulation. Which statistic is used decides the answer, and the point generalises: correlating priors with the terminal *level* of an accumulating variable recovers whichever coefficients set its drift, and drift coefficients are coded terms. Only a paired shock-minus-baseline difference isolates the causal response. Here the two disagree — level and response correlate at $r_s = -0.33$ across ensemble members.

Read on the level, tension is led by the base interest rate ($r_s = +0.40$) and equilibrium climate sensitivity ($r_s = +0.32$), with social priors at $|r_s| \approx 0.2$. Read on the response, a three-year stagflation pulse raises mean tension to a peak of $+0.0031$ five years later, positive in 76 % of prior draws, decaying back toward zero by year ten: a hump rather than a ramp, with transmission confirmed at lag zero and at multi-year lags (cumulative $+0.33$, within- R^2 0.20; Fig. 5; Morris screening (Morris, 1991; Campolongo et al., 2007) alongside). The largest response loading is a climate-damage parameter ($r_s = -0.35$), while the coded economy-to-society coefficient `SOCIAL_STRESS_UNEMPLOYMENT_SENS` is indistinguishable from zero on this statistic ($r_s = +0.13$): the channel reaching society is physico-economic and lagged, and it is not governed by the parameter whose name says it should be. The spectrum of Section 4 sees the same structure from the other side — its slow modes load on trust and tension.

Threshold-stepped sovereign cascades. The staircase shape is coded. What is not coded is *which* countries tip, in what order and at what dose: an oil-import burden propagates through reserves and debt issuance, leaving every importer untouched at $\leq 3\%$ of GDP, tipping three at 6 %, four at 10 %, twelve at 15 % and nineteen at 20 %. The ordering is not hand-set, but it is less fiscally determined than we previously reported. In the headline configuration the Spearman correlations of the tipping dose against pre-shock debt and reserves are -0.20 and -0.18 , against pre-shock GDP -0.38 : country size is the stronger single predictor and the fiscal margins are secondary. (The -0.46 and -0.44 quoted in an earlier draft come from a different initialisation than the tipping counts beside them; mixing the two was an error.) The defensible statement is that the ordering emerges from pre-shock state rather than a coded sequence, and turns on more than the debt trigger’s own variable — not that fiscal margins dominate it.

Power-law escalation of food-driven instability. The social increments are linear in their inputs (Appendix A), yet protest pressure responds to regional yield loss as a power law with exponent ≈ 1.09 — consistent in shape with the nonlinearity reported for food prices and unrest by Lagi et al. (2011) (a working paper; we cite it as such). Where the curvature originates matters, because composition across a block boundary and nonlinearity are separate properties. Across a sevenfold dose range the two legs carry similar exponents (1.17 for affordability stress, 1.09 for protest pressure) and the protest-to-affordability gain stays inside 0.16–0.22 with no systematic trend: the resource block generates the curvature and the social block transmits it. Cross-block composition brings the shock to society; it is not what makes the response nonlinear.

Cross-block synchrony of failures. On the deterministic 30-year baseline the count of active crisis states across the 57 agents rises from 5.9 per year in the first decade to 14.1 in the last. (Two crisis statistics appear in this paper and they are different objects: *active crisis states* summed over agents and years, as here, and crisis *episodes* per ensemble member, quoted in Section 4 as one to three sovereign episodes by 2053.) The natural summary of such a rise is a dispersion index, and it is the wrong one: a Fano index on a trending deterministic series measures the trend, so comparing it against the homogeneous-Poisson reference of unity tests nothing. Against the appropriate reference — an *inhomogeneous* Poisson surrogate carrying the same fitted intensity, 4,000 draws — the model shows no excess dispersion at all: surrogate mean 2.07 (5–95th 1.43–2.78) against a model value of 1.63, residual dispersion about a fitted trend 0.51. The counts are in fact *more* regular than Poisson. We note this because dispersion indices are routinely quoted for trending simulated series, where without a matched-intensity null they are uninformative.

The property that does hold is *synchrony across blocks*. Onsets of economic crises (sovereign debt, currency; $n = 68$) and of social crises (regime; $n = 85$) correlate at +0.52 at lag zero. Both series trend, so the raw correlation is confounded, and we give two detrended versions: after removing a linear trend from each the correlation is +0.60, and on Pearson residuals about a fitted log-linear intensity it is +0.56. Detrending therefore *strengthens* the association rather than explaining it away. Severing every cross-block channel — the parameter set is given in the archive — reduces the lag-zero correlation to +0.26 raw and +0.30 detrended, i.e. by half of its coupled value in both cases, while the rise in crisis counts falls only from 14.1 to 12.4 per year in the last decade. Correlation coefficients do not decompose additively into attributable shares, so we state the result as measured: removing the coupling halves the lag-zero correlation and leaves most of the trend intact. The two onset centroids are 2039.5 and 2043.7, four years apart — compatible with a lag-zero peak because the correlation is computed on annual counts whose year-to-year fluctuations co-move while their slow envelopes are offset. The ablation is what makes this a measurement rather than an interpretation.

Spatially structured risk. The diffusion itself is coded (neighbour spillover weight 0.05); what is measured is its effect. Switching it on costs under 0.01 % of the 2015–2023 retrospective fit, and it is not inert: switching the tension channel off raises the cross-country dispersion of tension by 6.5 % and quadrupling the weight lowers it by 2.2 %, monotonically in the weight, while the crisis count moves by up to 21 % when the whole spatial layer is removed. The *climate*-risk diffusion, by contrast, has no measurable downstream effect in the deterministic configuration — a limit of scope, not a defect. Climate reaches society only through discrete extreme events, which the deterministic runs switch off, leaving climate risk with only linear consumers, and a mean-preserving diffusion of a linear function changes nothing in aggregate. With extreme events enabled over a 12-seed ensemble the same ablation does move tension and crisis counts (1.7 % and 4.3 %). We do not claim the emergent spatial autocorrelation matches the observed record; that comparison lives in the archive’s geographic validation materials.

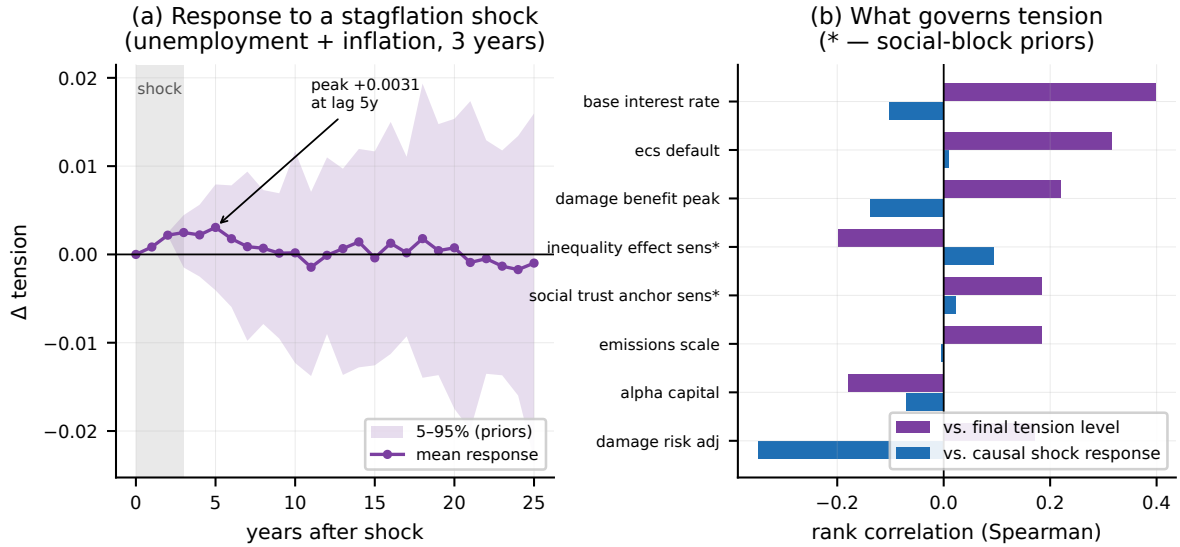


Figure 5: The economy-to-society channel. (a) Impulse response of mean tension to a three-year stagflation shock: a hump peaking around year five and decaying by year ten (5–95th band over priors). (b) What governs tension, contrasting the two questions: rank correlation of each prior with the final tension *level* and with the causal *shock response*. Social-block priors are marked *. The level is led by physico-economic priors, and the two statistics correlate at only -0.33 across members, which is why the claim rests on the paired response in (a) rather than on the level.

7 The instrument: ensembles, weak signals, and an adversarial tail sampler

The model is operated as an instrument with three modes, in increasing order of adversarialness.

7.1 Ensembles and scenario differences

The core is deterministic, so uncertainty is explicit: 500-member ensembles over the priors yield outcome distributions, and scenarios are compared as distributions against a common baseline. The whole workflow runs on a laptop (Apple M4 Pro: a world-year in 30–50 ms, the 500×30 -year ensemble in 109 s, the full 592-test suite in about two minutes), which is what makes ensemble-first practice a default rather than a special exercise. A worked case: a $\$50/\text{tCO}_2$ carbon tax cuts emissions by $\approx 5.5\%$ over ten years while raising tension and lowering trust through the energy-price-inflation channel — the political-economy cost invisible to single-domain models, monotone in the rate, and directly relevant to observed reversals (France 2018, Australia 2014); the full dose-response harness is Appendix B.

We recommend reading GIM in scenario *differences* rather than absolute levels, because the anchor uncertainties that dominate levels largely cancel between two runs sharing them. That is a design argument, not a demonstrated result: running one policy scenario across all 500 members and showing that the sign of the difference is stable is the second item of future work (Section 8), and until it is done the recommendation describes how the model is built, not how it behaves.

7.2 Weak signals

The joint state is monitored for early structure: Mahalanobis distance of the stationary increments from their recent baseline (Mahalanobis, 1936; Wilson and Hilferty, 1931), information-criterion change-point detection (Schwarz, 1978), and critical-slowness indicators (Scheffer et al., 2009). An injected 15 % output drop is flagged at the correct step while calm runs stay near the false-alarm rate. We report this in one paragraph rather than as a result: detecting a contemporaneous injected shock is change detection, not early warning before a transition, and a proper evaluation would need per-detector thresholds, detection delay, false-positive rate, sensitivity/specificity and a benchmark. We have not done that; the machinery ships as a monitoring aid and should not be cited as validated early-warning capability.

7.3 The game as an adversarial tail sampler

The same 57-actor core is wrapped into a single-player decision game (GODMODE): annual turns from 2023 to 2050 (27 turns), 158 dilemma cards compiled into typed policy actions, and seven catastrophe gates (temperature, biosphere, stability, economic and geopolitical strain, energy and food prices); losing any gate ends the run, reaching 2050 wins. A game simulates in about a second, and because players and scripted archetypes push the world into corners smooth history never visits, the layer functions as *adversarial stress testing* — targeted exploration of extreme regions of the state space. We previously called it an importance sampler; that term is wrong without a proposal density and explicit weights, and we withdraw it. Nothing below is a probability of a world event: these are frequencies conditional on the selected strategies and the game’s rules, and we report them as a stress-testing method rather than as a third scientific pillar.

A tail census. In the 1,002-game reference batch, survival is 23.5 %, losses spread over five causes (35 % climate, 22 % economic, 21 % geopolitical, 13 % stability, 8 % energy price), and 77 % of failures land in **2040–2049** (Fig. 6) — the same clock Section 6 sees in the crisis counts. Telemetry from the public deployment reproduces that shape with humans in the loop: of 401 runs started, 200 completed, 57 of them surviving (28.5 % of completers, 95 % CI ± 6.3 points; 14.2 % of starts). Under the assumption that runs going badly are abandoned more often, the completer rate is an upper bound. Human losses spread more broadly (31 % climate, 29 % economic, 15 % geopolitical, plus biosphere and food-price tails the bots rarely reach) and 90 % fall after 2040. This is *not* independent validation: players drive the same engine, so bot–human agreement shows robustness to how the strategy space is sampled, not corroboration against observation. No doctrine dominates — random play survives most often (28.4 %) against expansion (21.9 %) and decarbonisation (20.1 %), and each dies of its characteristic disease (green of economic strain, 43 % of its losses; growth of climate and stability, 39 % and 34 %). Single-axis optimisation shifting the binding constraint onto another subsystem is the game-mechanical face of the coupling this paper is about.

A diagnostic that found real defects. Adversarial play surfaced two symptoms that no 2015–2023 backtest could show: a resource price that never moved off its clamp, and government trust falling on hawk and dove paths alike — an attractor driven by an inequality drag $\approx 40\times$ the only restoring term, in the same socio-fiscal subspace the spectrum of Section 4 flags. Diagnosis into root causes came afterwards, from the price-validation

harness of Section 5, which found three separate accounting and buffer defects behind the pinned prices. Both fixes shipped as equilibrium anchors with the goldens bit-identical. The epistemic status of the layer is bounded — no goodness-of-fit against history, self-selected players, and difficulty gates tuned against the bot batches — but it explores exactly the regions that validation against smooth history cannot, and we regard it as part of the model’s quality programme rather than as evidence about the world.

Country policies can also be set by language-model agents (intentions compiled into the same typed actions, every consequence computed by the deterministic core, off in all validated configurations, with full audit logs). The whole environment ships as an open, pip-installable codebase with pinned goldens and manifest-bound runs (Sotiriadi and Osipov, 2026).

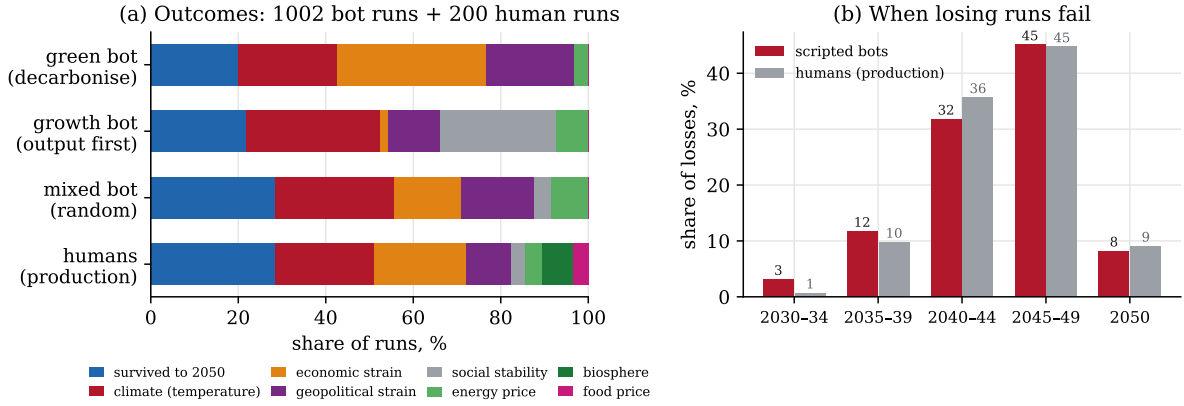


Figure 6: The tail census: 1,002 scripted playthroughs and 200 completed human runs (of 401 started; completer statistics shown, an upper bound under the stated attrition assumption), 2023–2050. (a) Outcome composition by archetype and for human players: no doctrine dominates, each dies of its characteristic cause, and the human loss spectrum is broader (biosphere and food-price tails the bots rarely hit). (b) Timing of failures, bots vs. humans: both distributions peak in 2040–2049 as slow accumulators come due.

8 Limitations

Boundedness is engineered and measured, not proven. The anchors that bound the loop could mask genuine instabilities of the real system; no formal stability proof for the full nonlinear map exists, and the evidence of Section 4 is local, numerical, and computed on a 408-dimensional projection of a larger persistent state. The trust anchor is the clearest trade-off: without it the model has a unit root in an institutional variable and the spectral radius reflects that defect rather than the coupling; with it the model reverts toward each agent’s base-year level, which is a non-neutral assumption about the world over thirty years. Its mean-reversion pull was moreover selected on the very diagnostic the section reports, and we give the spectra at four settings so the reader can price it.

Structural uncertainty is not represented in the ensembles, and it is large. The 500-member fan varies parameters. It does not vary structure, initial conditions, or scenario inputs, and the paper itself contains evidence that structural uncertainty is comparable to or larger than the parametric kind: removing the SSP2 productivity anchor moves 2100 warming from 2.75 to about 2°C, far outside the parametric fan, and renormalising

production to constant returns moves 2100 world product by 7–9%, of the same order as the entire 30-year parametric spread. A reader should decompose $U_{\text{total}} = U_{\text{parameter}} + U_{\text{initial state}} + U_{\text{structural}} + U_{\text{scenario}}$ and treat this paper as quantifying only the first.

The parameter set mixes three provenances — literature-grounded, calibrated against the 2015–2023 backtest, and expert-specified with no external source; Tables A1 and A2 mark each, and describing all 33 as “literature-grounded” would be wrong. Three (the warming-benefit cap, the growth-damage coefficient and the CH₄ feedback) are zero in the headline configuration and sampled only in tail scenarios, so the headline ensemble varies 30 of the 33 about non-zero centres.

Resource prices reach the economy unevenly. Forcing each price by a fixed multiple over a ten-year run moves world output by -5.5% (energy), -7.9% (food) and $+0.8\%$ (metals) at a fivefold shock. The metals price is close to inert downstream. Food reaches society mainly through the trade balance and the sovereign channel rather than through consumer prices, and the food cost-push term into inflation was added only in this version.

Climate reaches society only through discrete events. There is no continuous climate-to-tension term in the headline configuration. One exists in the code, disabled: the canonical prior for it (Hsiang et al., 2013) reports a 14% rise in intergroup conflict per standard deviation of warming, but that meta-analysis is contested on sample selection and analytical coherence (Buhaug et al., 2014), with a reply disputing the reanalysis. The direction is defensible and the magnitude is not identified, so we ship the mechanism off and tagged rather than choosing a number, and every result in this paper is produced with it disabled.

Crop response to warming is newly endogenous and is not adaptation. Before this version warming did not affect crops at all. It now does, at the per-degree yield losses of Zhao et al. (2017) composed with an agricultural supply price elasticity (Haile et al., 2016); either alone is misleading, since the yield channel without a supply response drives the food price to its ceiling. But that supply response is a *price-induced acreage and output adjustment* — a market mechanism partly offsetting a yield shock, not climate adaptation, which would mean crop switching, shifted sowing dates, irrigation, cultivars, relocation and technology. GIM has none of those, and we withdraw the earlier description of the pair as supplying adaptation. It gives a $+24\%$ food price by 2053, inside the IPCC-assessed range for climate-driven cereal prices at 2050, an anchor it was not fitted to.

Expectations are adaptive, and long-run growth is weakly identified. Agents look at the recent past; a tested model-consistent expectations mechanism is off by default (its marginal effect proved unstable), so anticipatory responses to announced policy are absent. Forward projections anchor productivity drift to an external scenario, and the baseline compounds at $3.06\%/yr$ to 2053 but $4.67\%/yr$ thereafter — an artefact of extrapolating the R&D and convergence machinery well past its calibration window, not a projection. Absolute long-horizon levels inherit this, which is why scenario differences are the recommended mode.

The climate block is a fast emulator, not an Earth-system model; the strongest carbon feedbacks are deeply uncertain, switchable and relegated to tail scenarios. Non-CO₂ forcing is exogenous, from the SSP2-4.5 marker (Meinshausen et al., 2020): the loop is closed for CO₂ and open for everything else, which qualifies the “closed loop” description. The radiative-forcing coefficient is the Myhre et al. (1998) value, not the AR6 update (Forster et al., 2021) — because sensitivity is calibrated at doubling the equilibrium anchor is unaffected and the transient response is checked directly, but a reader comparing GIM’s forcing to AR6 will find the older constant. Damage estimates are carried as an interval.

The socio-political block mixes anchored and expert-set pieces. Substantial parts are tied to literature (war-size tail, migration gravity, misery-index ratio, governance anchors); coupling thresholds and cross-border tension diffusion remain expert priors, tagged in the registry. The block supports relative risk ranking (AUC ≈ 0.74 ; vintage-clean 0.84) — and, as Section 5 reports, no better than its own strongest input — not event prediction. The persistent social mode of Section 4 means socio-political initial-state errors do not wash out, a larger inherited uncertainty than the economic case.

No nominal exchange rate, and the currency channel depends on that. The currency-crisis mechanism fires at a rate and duration consistent with the observed record (3.5 % of eligible agent-years against 3–5 % (Laeven and Valencia, 2020); mean duration 1.9 years against 1–3), but only because episode length is bounded explicitly. Nothing rebuilds reserves, since there is no depreciation to compress imports, so a reserve-accumulation identity is implemented and deliberately disabled: switching it on drives the crisis rate to 8 %. Because reserves and debt drive both the sovereign channel and the late-game failures of Section 7, the post-2040 rise in active crises and the concentration of game losses in 2040–2049 may be partly an artefact of this incomplete subsystem; a sensitivity run with the identity enabled is the third item of future work. Agents that do not issue their own currency, or that issue a reserve currency, are exempt from the trigger by an explicit list, because the model has no monetary-union concept and would otherwise flag euro-area members — as it did, until the historical backtest caught it.

Aggregation. Resources are aggregate (no sectoral energy technologies, no endogenous renewable learning curves); there is no portfolio capital account and no pandemic submodel (epidemic shocks are extreme events); state-dominated political economies use shared functional forms with country-specific states; and the annual map is evaluated once, so within-year equilibration is imposed rather than converged to.

9 Conclusions

GIM v20.1.0 is a closed seven-domain, 57-agent world simulator whose behaviour we measure rather than assert. (i) It stays together: a spectral radius of 1.036–1.100 whose above-unity modes are mainly economic, whose climate coordinates carry none of the above-unity mass while containing a conservative direction by construction, with contracting economic and climate perturbations and near-linear growth of parametric spread over thirty years — bounded over the tested horizon, not asymptotically stable. (ii) It is anchored: in-sample and free-running held-out retrospective evaluation across output, emissions, temperature and resource prices, external cost-of-carbon cross-checks, and an input-vintage-robust conflict ranking reported honestly against its own best single covariate, which it does not beat.

(iii) It is responsive: a lagged social impulse response, threshold cascades whose ordering is endogenous, power-law escalation whose curvature we trace to the block that generates it, and failures that synchronise across blocks more than the blocks alone would produce — a synchrony that survives detrending.

Underneath is a stated philosophy: the deterministic core encodes physical conservation laws, accounting identities and slow empirical regularities with explicit restoring forces, and parametric uncertainty is carried by ensembles over their strengths. Around it is an instrument: prior ensembles, weak-signal detectors, and a game layer that adversarially samples the tails and has already paid for itself by exposing two structural degeneracies no retrospective test could see. Two of the paper’s own diagnostics returned negative results — the history-matching filter does not bind, and the conflict composite does not beat its strongest input — and we report both rather than removing the tests that produced them.

The model does not forecast; it maps the consequences of assumptions, the distribution of risks, and the ways a managed world can fail — comparatively, reproducibly, and on a laptop.

Code and data availability

The complete model, calibration data with checksums, every validation and stability harness cited in the text, the game backend, and the figure-generation scripts are archived as GIM v20.1.0 on Zenodo at <https://doi.org/10.5281/zenodo.22102520> (Sotiriadi and Osiptsov, 2026). That is the version DOI of the frozen deposit from which every number in this paper was produced, and it is the record we ask readers to cite; the concept DOI <https://doi.org/10.5281/zenodo.21575176> resolves to whichever version is current and should be used only when referring to the model in general rather than to these results. The archive is the citable record; the development repository at <https://github.com/Theclimateguy/GIM> (tag v20.1.0) is a convenience mirror and is *not* the archival copy. All code and configuration are released under the Apache License 2.0, an OSI-approved licence. The archive contains the model source; the user manual and command reference; every configuration file and scenario definition; the pinned observed fixtures; the pre-processing, run-control and post-processing scripts for every number, table and figure in this paper; and the 592-test regression suite with its golden trajectories. Reproducing a result is one command per harness, listed in the manual; the figure scripts are under `Paper/figures/` and the experiment scripts under `Paper/revision/scripts/`.

Input and comparison datasets are third-party and are handled as follows.

Table 4: Input and evaluation datasets: version, role and redistribution status. “Mirrored” means the file is inside the frozen archive; “script” means the archive carries a download script and a checksum but not the data, because the licence does not permit redistribution.

Dataset	Version / vintage	Role	Status
World Bank WDI (World Bank, 2024b)	1990–2024 pull, 2026-06	output, population, debt, reserves, military expenditure, Gini	CC BY 4.0, mirrored
World Bank WGI (Kaufmann et al., 2011)	1990–2024, source 57	regime stability, trust, legitimacy anchors	CC BY 4.0, mirrored

Table 4 — continued

Dataset	Version / vintage	Role	Status
Penn World Table (Feenstra et al., 2015)	10.01	factor shares, depreciation	mirrored
Global Carbon Budget (Friedlingstein et al., 2023)	GCB 2024 workbook, fossil excl. carbonation	emission record and target series	CC BY 4.0, mirrored
NOAA GML global mean CO ₂ (Lan et al., 2024)	global annual mean series, 2024	base-year atmospheric stock	public domain, mirrored
HadCRUT (Morice et al., 2021)	5.1.0.0, rebased to 1850–1900	temperature target series, 1990–2023 and 2015–2023	Open Government Licence, mirrored
RCMIP non-CO ₂ forcing (Meinshausen et al., 2020)	SSP2-4.5 marker	exogenous non-CO ₂ forcing path	mirrored
World Bank Pink Sheet (World Bank, 2024a)	annual nominal indices, 2010=100, retrieved 2026-08-24	resource-price target series	CC BY 4.0, mirrored
UCDP/PRIO ACD (Gleditsch et al., 2002; Davies et al., 2024)	v24.1	conflict ranking target	CC BY 4.0, mirrored
SIPRI Military Expenditure (SIPRI, 2024)	2023 edition	CINC military component	non-commercial use only — derived 57-actor aggregate mirrored, source workbook by script
SWIID (Solt, 2020)	v9.x	inequality anchor	script
Correlates of War NMC	v6.0	CINC validation	script

Table 4 — continued

Dataset	Version / vintage	Role	Status
Hofstede cultural dimensions (Hofstede et al., 2010)	2015 release	three dimensions (PDI, IDV, UAI) enter the tension and trust equations	source table not redistributed: the scores are not published under an open licence. The public archive carries the ingestion script and a checksum of the expected input, together with the three-dimension vector for the 57 modelled agents that the engine consumes; the full multi-country source table must be obtained from the rights holder

Author contributions

N.S.S.: conceptualization, methodology, software (the simulation engine and the game layer), investigation, writing — original draft. A.A.O.: validation (model validation programme), writing — original draft and review & editing. A.N.K.: validation (model and game-simulator testing), formal analysis (assessment of results), writing — review & editing. The archived software is authored by N.S.S. and A.A.O.; the Zenodo deposit’s author list reflects the software, the byline reflects the paper.

Competing interests

N.S.S. and A.A.O. are employees of JSC Sberbank of Russia, whose risk-management activities are adjacent to the model’s stated use cases; the bank had no role in study design, analysis, or the decision to publish. The authors declare no other competing interests.

Ethics and player telemetry

The gameplay telemetry reported in Section 7 contains no personal data. The public deployment records, per run, only parameters of the run itself: a randomly generated run identifier, the random seed, the in-game outcome (win/loss, cause, year), the composite score, and final in-game aggregate metrics. There are no accounts. No names, e-mail addresses, IP addresses, demographic attributes, device identifiers or any other information that could identify or be linked to a player is collected, stored or derived at any point; an optional nickname may be submitted after a run for the public leaderboard and is excluded from the analysis reported here. Nothing in the collection is personalised, so no individual-level data exist to report and none are reported. No ethics review was sought, and we state plainly that none has been obtained rather than asserting that any body has adjudicated the question.

Financial support

This research received no funding of any kind and was carried out entirely on the authors' own time. Article processing charges are borne by the authors.

Use of generative-AI tools

Large-language-model tools were used in the preparation of this manuscript. Specifically: Anthropic Claude and OpenAI GPT models were used (i) as coding assistants in the development of the model, its harnesses and its figure scripts; (ii) to check the internal numerical consistency of the manuscript against the harness outputs; (iii) to format and check bibliographic entries; and (iv) to review draft text for clarity, grammar and structure, and to propose rewordings of passages the authors then wrote or rewrote themselves. No section of this manuscript was generated by a language model and adopted without the authors rewriting it, and no analysis, result, interpretation or scientific claim was produced by a language model: every number reported here comes from the deterministic model harnesses named in the text and was verified by the authors, who take full responsibility for the content. LLM use *inside* the model is a documented, optional feature (Section 7) and is off in every configuration reported in this paper.

Acknowledgements

TODO: add, or delete this section. We thank the external players whose anonymous gameplay produced the telemetry of Section 7.

A Model specification

Symbols follow Algorithm 1; agent subscripts are dropped where unambiguous. Every constant quoted here is the headline value in `gim/core/calibration_params.py` of release v20.1.0 and is tagged there with its provenance.

A.1 Production, capital and productivity

Output is nested CES in capital, labour and energy, normalized to Cobb–Douglas at each country’s base point (Arrow et al., 1961):

$$Y = A \Theta \text{KE}_0^{\alpha+\gamma} \left(a_K (K/K_0)^{\rho_{KE}} + a_E (E/E_0)^{\rho_{KE}} \right)^{(\alpha+\gamma)/\rho_{KE}} L^\beta, \quad \rho_{KE} = \frac{\sigma_{KE}-1}{\sigma_{KE}}, \quad (\text{A1})$$

with $\alpha = 0.30$, $\beta = 0.60$, $\gamma = 0.042$, $a_K = \alpha/(\alpha + \gamma)$, $a_E = \gamma/(\alpha + \gamma)$, $\text{KE}_0 = K_0^{a_K} E_0^{a_E}$. At the base point the inner aggregator equals one because $a_K + a_E = 1$, so levels and first derivatives reproduce the Cobb–Douglas limit for any σ_{KE} , while substitution away from it is governed by $\sigma_{KE} = 0.4$ (gross complementarity (van der Werf, 2008; Koetse et al., 2008); factor shares from the Penn World Table (Feenstra et al., 2015; Gollin, 2002)). The exponent sum 0.942 encodes mild decreasing returns; per-country scale factors A pin base-year levels, so renormalising to constant returns moves 2100 world product by 7–9 %. Θ is the climate damage multiplier of Eq. A8. Capital accumulates as

$$K_{t+1} = (1 - \delta)K_t + s_t Y_t, \quad \delta = 0.05, \quad (\text{A2})$$

with $s_t \approx 0.24$ modulated by political stability and social tension (Mody et al., 2012). Productivity grows at $\approx 1\%$ /yr baseline with an R&D stock à la Jones (1995), trade-mediated diffusion, and β -convergence toward the frontier at slope 0.0093 per log income gap, validated on the cross-section.

A.2 Labour market and prices

Unemployment partially adjusts toward an Okun target (Okun, 1962):

$$u_t^* = \bar{u} - \omega (g_t - g^*), \quad u_{t+1} = u_t + \phi_u (u_t^* - u_t), \quad (\text{A3})$$

with NAIRU $\bar{u} = 0.045$, Okun coefficient $\omega = 0.30$, potential growth $g^* = 0.025$, adjustment speed $\phi_u = 0.5$, and u clamped to $[0.01, 0.35]$. Inflation follows an expectations-augmented flat Phillips curve (Phillips, 1958) with energy and food cost-push and an optional quantity-theory term:

$$\pi_{t+1} = \underbrace{\theta \pi^* + (1 - \theta) \pi_t}_{\pi^e} + \kappa_\pi (\bar{u} - u_{t+1}) + c_E \hat{p}_t^E + c_F \hat{p}_t^F + \lambda_M \max(0, \widehat{M}_t - g^* - \pi^*), \quad (\text{A4})$$

with anchor weight $\theta = 0.5$, target $\pi^* = 0.02$, slope $\kappa_\pi = 0.25$, energy pass-through $c_E = 0.05$, food pass-through $c_F = 0.04$, money pass-through $\lambda_M = 0.027$, hats denoting proportional changes, and π clamped to $[-0.02, 0.30]$. Policy rates follow country Taylor rules capped at ± 6 pp (Taylor, 1993).

A.3 Public finance, banking and the sovereign channel

Government budgets close each year: the primary balance plus interest on the debt stock determines new issuance, the effective interest rate rises with the debt ratio (a financial accelerator (Bernanke et al., 1999)), and credit ratings are a function of debt, growth, inflation and stability. The financial side is stock-flow consistent in the sense of Godley and Lavoie (2007): within GIM’s simplified banking balance sheet each loan creates a matching deposit and the money stock is identically deposits plus loans; there is no interbank market, no portfolio choice and no nominal exchange rate. Bilateral trade settles in reserves; world net exports are demeaned pro rata to GDP so that the world closes to 4×10^{-18} of world product. Trade volumes follow a gravity law in distance and size (Head and Mayer, 2014).

A.4 Resources and prices

Three aggregate resources (energy, food, metals) carry stocks, annual supply, demand and a world clearing price index normalized to 1.0 in the base year. Demand grows with population and income (income elasticity 0.50 for energy) and responds to price at the market demand elasticity 0.40 (Labandeira et al., 2017). Supply responds to price and to stock depletion; food supply additionally responds at the acreage elasticity 0.143 (Haile et al., 2016), and food yields fall by 4.9% per °C of warming (Zhao et al., 2017). Each price mean-reverts in log space toward its reference:

$$\ln p_{t+1} \leftarrow \ln p_{t+1} + \psi (\ln p^{\text{ref}} - \ln p_{t+1}), \quad \psi = 0.15 \text{ yr}^{-1}. \quad (\text{A5})$$

A.5 Climate and carbon

Four-pool AR6 impulse response (Joos et al., 2013), integrated exactly over the annual step:

$$C_{t+1}^{(i)} = C_t^{(i)} e^{-1/\tau_i} + a_i E_t, \quad a = (0.2173, 0.2240, 0.2824, 0.2763), \quad \tau = (\infty, 394, 36.5, 4.3) \text{ yr}, \quad (\text{A6})$$

with $\sum_i a_i = 1$ exactly (shares rounded here). The atmospheric stock is $C = C_{\text{pre}} + \sum_i C^{(i)}$ with $C_{\text{pre}} = 2184 \text{ GtCO}_2$; note that $\tau_1 = \infty$ makes the first pool a perfect integrator, which is the conservative direction Section 4 discusses. Forcing is $F = 5.35 \ln(C/C_0) + F_{\text{non-CO}_2}$ — the Myhre et al. (1998) coefficient, implying $F_{2\times} = 3.71 \text{ W m}^{-2}$ and, at the calibrated equilibrium sensitivity of 3.0°C , a feedback parameter $\lambda = F_{2\times}/\text{ECS} = 1.24 \text{ W m}^{-2} \text{ K}^{-1}$; the AR6-updated $F_{2\times}$ is higher (Forster et al., 2021), but since sensitivity is calibrated at doubling the equilibrium anchor is unaffected and the transient response is checked directly (TCR 1.79°C , within the AR6 likely range). $F_{\text{non-CO}_2}$ is exogenous, taken from the SSP2-4.5 marker (Meinshausen et al., 2020). Two-layer heat balance (Geoffroy et al., 2013), explicit Euler at $\Delta t = 1 \text{ yr}$:

$$C_s \dot{T} = F - \lambda T - \kappa(T - T_d), \quad C_d \dot{T}_d = \kappa(T - T_d), \quad (\text{A7})$$

with $C_s = 8$, $C_d = 100 \text{ W yr m}^{-2} \text{ K}^{-1}$ and $\kappa = 1.0 \text{ W m}^{-2} \text{ K}^{-1}$.

Damage enters production as a multiplier, renormalized so that the base year is reproduced exactly:

$$\Theta(T) = \frac{1 - \zeta T^2}{1 - \zeta T_{2023}^2}, \quad \zeta = 0.0078, \quad (\text{A8})$$

so $\Theta(T_{2023}) = 1$ by construction. The unnormalized function $1 - \zeta T^2$ gives 3.1, 7.0 and 12.5% losses at +2, +3 and +4 °C relative to preindustrial, anchored at the Howard–Stern

productivity-corrected central estimate (Howard and Sterner, 2017; Nordhaus, 2017); a Burke-type growth-damage channel is switchable and off in the headline configuration (Burke et al., 2015, 2023).

A.6 Society, politics and discrete crises

Institutional trust is a $[0, 1]$ state driven by deviations, not levels — the distinction Section 4 explains:

$$\begin{aligned}\Delta \text{tr}_t &= \eta_y \frac{y_t - y^{\text{ref}}}{y^{\text{ref}}} + \eta_u (u_t - \bar{u}) + \eta_\pi (\pi_t - \pi^*) + \eta_G (G_t - G^{\text{ref}}) + \eta_S \max(0, S_t - S^{\text{thr}}), \\ \text{tr}_{t+1} &= \text{clip}_{[0,1]} \left(\text{tr}_t + \Delta \text{tr}_t + \underbrace{\chi (\text{tr}^{\text{anchor}} - \text{tr}_t - \Delta \text{tr}_t)}_{\text{mean reversion, } \chi=0.10 \text{ yr}^{-1}} \right),\end{aligned}\tag{A9}$$

with $\eta_y = 5 \times 10^{-5}$ per unit of relative income gap, $\eta_u = -0.030$, $\eta_\pi = -0.015$ (the two-to-one misery-index ratio (Di Tella et al., 2001)), $\eta_G = -0.0004$ per Gini point (Wilkinson and Pickett, 2009), $\eta_S = -0.08$ above a tension threshold $S^{\text{thr}} = 0.30$, and $\text{tr}^{\text{anchor}}$ each agent’s own base-year WGI value. Social tension $S \in [0, 1]$ accumulates:

$$\begin{aligned}S_{t+1} &= \text{clip}_{[0,1]} \left(S_t + \underbrace{\sigma_u (u_t - \bar{u}) + \sigma_\pi (\pi_t - \pi^*) + \sigma_G (G_t - G^{\text{ref}})}_{\text{economic stress}} (1 - \text{IDV}/100) \right. \\ &\quad \left. + \underbrace{\sigma_{\text{tr}} (\text{tr}^{\text{ref}} - \text{tr}_{t+1})}_{\text{trust-gap term, } \sigma_{\text{tr}}=0.060} + w_{\text{sp}} (\bar{S}^{\text{N}} - S_t) \right),\end{aligned}\tag{A10}$$

with $\sigma_u = 0.010$ per point of unemployment, $\sigma_\pi = 0.005$ per point of inflation, $\sigma_G = 5 \times 10^{-4}$ damped by the Hofstede individualism score, and a neighbour spillover $w_{\text{sp}} = 0.05$ toward the geographic-neighbourhood mean $\bar{S}^{\text{N}}$. Two distinct mechanisms carry the word “anchor” in earlier versions of this paper and we now name them separately: the *trust mean-reversion pull* $\chi = 0.10 \text{ yr}^{-1}$ in Eq. A9, and the *trust-gap effect on tension* $\sigma_{\text{tr}} = 0.060$ in Eq. A10. Both are on in every result reported here (Table 2). Legitimacy is $0.6 \text{ tr} + 0.4(1 - S)$ and sets the policy-space multiplier. Protest pressure and a food-affordability index are diagnostic composites of tension, inequality and the food price.

Discrete crises are a state machine with expert-set thresholds, tagged as such in the registry:

Sovereign debt. Trigger: debt $> 120\%$ of GDP *and* effective rate $> 12\%$. Effect: debt $\times 0.60$ (a 40% write-down), output $\times 0.93$, unemployment $+ 5 \text{ pp}$ (capped at 30%), trust $- 0.15$, tension $+ 0.15$, stability $- 0.15$. Exit: debt below 70% and rate below 8%, within at most six years.

Currency. Trigger: external debt $> 50\%$ of GDP *and* current-account deficit $> 4\%$ *and* reserve cover < 3 months. Effect: debt $\times 1.10$, output $\times 0.85$, unemployment $+ 6 \text{ pp}$. Exit: bounded episode length. Agents that do not issue their own currency or that issue a reserve currency are exempt by an explicit list (Section 8).

Regime collapse. Trigger: trust < 0.20 *and* tension > 0.80 . Effect: output $\times 0.8$, capital $\times 0.7$ (Barro and Ursúa, 2008). Exit: trust above 0.25 and tension below 0.60.

Crisis severity is optionally heavy-tailed with a war-size exponent $\alpha_w \approx 1.5$ (Richardson, 1948; Clauset et al., 2009).

A.7 Geopolitics, demography and migration

Interstate relations are a bilateral matrix updated by alliance, sanction, trade and conflict events; a CINC-style capability index (Singer et al., 1972) is reconstructed from population, energy use, output and military expenditure (SIPRI, 2024) and validated against published CINC values. Population evolves from base birth and death rates (0.025, 0.012/yr) modulated by income and crisis state; labour supply is a fixed participation share of population. Migration flows follow a gravity form with income and conflict push terms, capped at a small share of population per year. A classic Richardson arms-race term was tested and rejected on the 1995–2024 military-expenditure panel ($p = 0.74$) and is not in the model.

A.8 Prior specification, history matching and ensemble formation

Priors are per-parameter triangular (30 of 33), lognormal (2) or normal (1) over the ranges of Tables A1–A2; the full registry of 287 parameters with calibration-status tags is in the archive. Implausibility uses tolerances tol_o of 6 trillion USD (world product), 2 Gt (emissions) and 0.15°C (temperature) — observational uncertainty bundled with an explicit model-discrepancy allowance — with the cut $\max_o I_o \leq 3$. Resource-price tolerances are set instead to the error a flat price makes, so an implausibility below one beats the null. The published configuration uses a single wave of 60 Latin-style random draws over the six backtest-driving parameters; at the three-sigma cut the acceptance rate is 100 %, so the not-ruled-out-yet region coincides with the prior box and ensembles draw all 33 parameters independently from their priors (Section 4).

Table A1: Key priors, part 1: the 19 parameters discussed in the text (units inline; dimensionless unless stated). “Provenance” is L = literature, B = calibrated against the 2015–2023 backtest, E = expert-specified, N = numerical regularisation — the three-way distinction Section 8 insists on.

Parameter	Central	Range	Prov.	Source
Equilibrium climate sensitivity, $^\circ\text{C}$	3.0	1.5–6.0	L	(IPCC, 2021; Otto et al., 2013; Sherwood et al., 2020)
Surface heat capacity, $\text{W yr m}^{-2}\text{K}^{-1}$	8.0	6.0–18.0	L/B	(Geoffroy et al., 2013)
Ocean heat exchange, $\text{W m}^{-2}\text{K}^{-1}$	1.0	0.4–1.3	L/B	(Geoffroy et al., 2013)
Emissions scale	0.98	0.90–1.05	L	(Friedlingstein et al., 2023)
Structural decarbonization rate, yr^{-1}	0.016	0.010–0.030	B	GCP/WDI trend (Friedlingstein et al., 2023); 2015–2023 backtest
Capital share α	0.30	0.22–0.40	L	(Feenstra et al., 2015; Gollin, 2002)
Labor share β	0.60	0.50–0.72	L	(Feenstra et al., 2015; Gollin, 2002)
Energy elasticity γ	0.042	0.015–0.080	B	2015–2023 backtest
Capital–energy substitution σ_{KE}	0.40	0.20–0.60	L	(van der Werf, 2008; Koetse et al., 2008)
Depreciation rate δ , yr^{-1}	0.05	0.03–0.08	L	(Feenstra et al., 2015)
TFP sensitivity to R&D	0.30	0.10–0.60	B	calibration to backtest

Table A1 — continued

Parameter	Central	Range	Prov.	Source
Damage coefficient ζ	0.0078	0.0015–0.025	L	(Howard and Sterner, 2017; Nordhaus, 2017; Burke et al., 2015)
Pure time preference	0.015	0.001–0.030	L	(Nordhaus, 2017; Stern, 2007)
Elasticity of marginal utility η	1.45	1.0–2.0	L	(Nordhaus, 2017; Stern, 2007)
Market demand elasticity	0.40	0.20–0.70	L	(Labandeira et al., 2017)
<i>Social block (validated by sign and channel, Sec. 6)</i>				
Tension stress from unemployment σ_u	0.010	0.007–0.014	L/E	(Di Tella et al., 2001); structural prior
Tension stress from inflation σ_π	0.005	0.003–0.007	L	(Di Tella et al., 2001), aversion $\approx 2:1$
Inequality effect on tension σ_G	0.0005	0.0003–0.0007	L/E	(Wilkinson and Pickett, 2009)
Trust-gap effect on tension σ_{tr} (Eq. A10)	0.060	0.039–0.081	B	calibrated to WGI trust persistence
Trust mean-reversion pull χ , yr^{-1} (Eq. A9)	0.10	0–0.20	N	weakest value lowering ρ at all three linearization points (Sec. 4)
<i>Resource block</i>				
Energy demand income elasticity	0.50	0.33–0.60	B	IEA intensity trend vs. world GDP growth
Food yield loss per $^\circ\text{C}$	0.049	0.031–0.074	L	(Zhao et al., 2017), four independent method families
Food supply price elasticity	0.143	0.045–0.79	L	(Haile et al., 2016), aggregate long-run growing area
Food price pass-through to CPI c_F	0.04	0.02–0.08	L	CPI food weight $\times$ commodity-to-retail pass-through

Table A2: Key priors, part 2: the remaining 14 of the 33 sampled parameters. “Headline” records whether the parameter is sampled about a non-zero centre in the 500-member headline ensemble or is fixed at zero there and sampled only in tail scenarios — the distinction Section 8 makes explicit.

Parameter	Central	Range	Prov.	Headline	Source
Warming-benefit cap	0	0–0.02	E	tails only	prior (CO_2 fertilisation)
Warming-benefit peak, $^\circ\text{C}$	0.3	0–1.0	E	inactive	deprecated while the cap is 0
Damage risk adjustment	0.005	0–0.02	E	sampled	prior

Table A2 — *continued*

Parameter	Central	Range	Prov.	Headline	Source
Deep-ocean heat capacity	100	40–200	L	sampled	(Geoffroy et al., 2013); DICE-2016 (Nordhaus, 2018)
Land-use CO ₂ , Gt/yr	0.6	0–1.5	L	sampled	(Friedlingstein et al., 2023)
Carbon feedback, GtCO ₂ /°C	1.5	0–5.0	L	sampled	AR6 WG1 Ch.5 (IPCC, 2021); permafrost literature
CH ₄ feedback, W m ^{−2} /°C	0.03	0–0.1	L	0 in head-line	AR6 WG1 Ch.5 (IPCC, 2021)
Base birth rate	0.025	0.015–0.035	L	sampled	(World Bank, 2024b)
Base death rate	0.012	0.007–0.017	L	sampled	(World Bank, 2024b)
Neutral interest rate	0.02	0.005–0.04	L	sampled	IMF WEO; r^* literature
Growth-damage TFP coefficient	0	0–0.001	L	0 in head-line	(Burke et al., 2015)
War-size tail exponent α_w	1.5	1.3–1.8	L	sampled	(Richardson, 1948; Clauset et al., 2009)
Migration base / max share	0.001/0.003	see registry	E	sampled	UN DESA; UNHCR
Regime-collapse output/capital mult.	0.8/0.7	0.7–0.9 / 0.55–0.85	L	sampled	(Barro and Ursúa, 2008)

B Characterisation of cross-sector transfer functions

For three shocks we take a published conclusion from a sectoral model where one exists, feed GIM the same exogenous shock, and ask whether it (a) reproduces the sectoral first-order effect and (b) exhibits a cross-sector consequence the sectoral model cannot represent by construction. Runs are deterministic (57 countries, ten years, extreme events off) and each scenario carries a dose–response sweep. We have retitled this appendix: an earlier version promised a “benchmark against a named sectoral model” in all three cases and delivered a named model in one. What it actually establishes is the *shape and gain* of three cross-sector transfer functions, and that is how we report it. Reporting a nonzero GIM slope against a sectoral anchor’s flat one would not be a measurement in any case: a model without a society block has zero slope on the society axis by construction.

Carbon tax (sectoral anchor: DICE-2016R (Nordhaus, 2017, 2018)). Fed the same \$50/tCO₂ tax, GIM reproduces the emissions decline ($\approx 5.5\%$ over ten years; log–log

exponent 0.88 in the tax rate, $R^2 = 0.999$ over the sweep — mildly saturating) and adds the political-economy channel absent from DICE: tension rises and trust falls in the rate, with exponent 1.82 ($R^2 = 0.97$). This is the one case where a published sectoral number is directly comparable: DICE’s optimal carbon price of $\approx \$31/\text{tCO}_2$ (2015, rising $\approx 3\%/yr$) is the anchor for the tax level, and the emissions response is the first-order effect both models share.

Oil shock (no named sectoral anchor). An oil-import burden propagates into sovereign finance through reserves and debt issuance. The response is threshold-stepped: no importer tips at 3 % of GDP per year, three tip at 6 %, four at 10 %, twelve at 15 % and nineteen at 20 % (Sri Lanka 2022 is the real-world instance of the mechanism). Two features of the construction bear stating and both weaken the experiment. First, the supply cut is converted into an import bill *outside* the model, using an assumed price elasticity (0.30) and import intensity (0.05) hard-coded in the benchmark harness, and injected directly into public debt and FX reserves. The scenario therefore exercises the finance-to-sovereign-crisis leg and does *not* test the resources-to-finance coupling inside GIM. Second, we cite no published sectoral conclusion here: “energy-system models” is not a named anchor, and we do not claim one.

Crop shock (assessment anchor: AgMIP/AR6 range). A regional yield loss raises food-affordability stress and protest pressure as power laws of similar exponent (1.17 and 1.09, $R^2 = 0.997$ and 0.976) (Lagi et al., 2011): the curvature is generated in the resource block and transmitted to society, with the protest-to-affordability gain staying inside 0.16–0.22 across a sevenfold dose range. The assessment chain stops at the food-security index.

The three channels take distinct forms — power-law, threshold-stepped, and power-law with a different exponent (Fig. B1) — and that variety, rather than a nonzero slope, is what the exercise establishes.

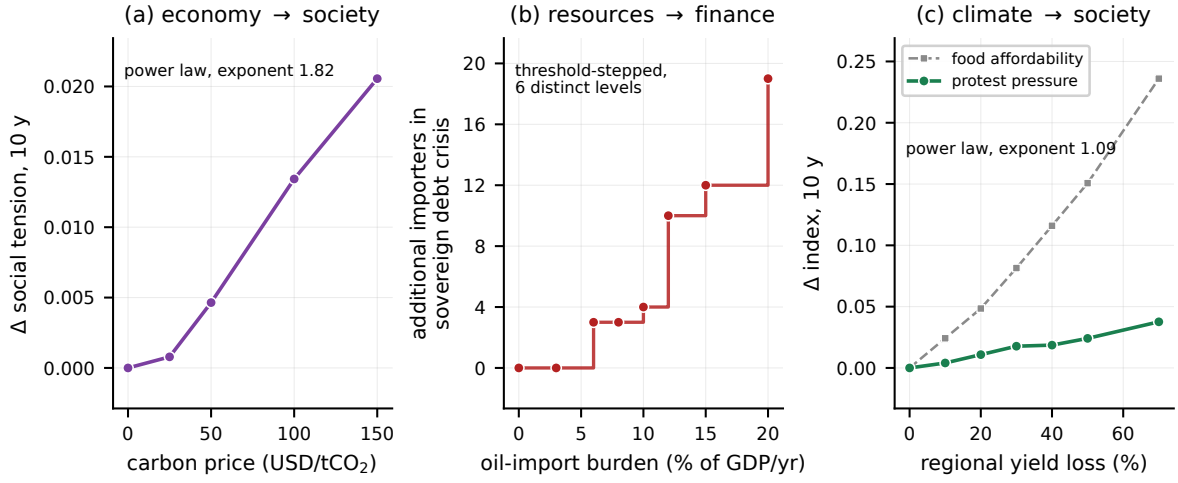


Figure B1: Three cross-sector transfer functions, each in its own units. (a) Carbon price to social tension: a power law, exponent 1.82. (b) Oil-import burden to the count of importers entering sovereign debt crisis: threshold-stepped, six distinct levels, no importer tipping below 6 % of GDP per year. (c) Regional yield loss to food affordability (grey) and protest pressure (green): power laws of similar exponent, 1.17 and 1.09, so the curvature is generated in the resource block and transmitted rather than compounded. Note that the panel's input is an exogenous *yield* loss, not a temperature change: this is a resources-to-society channel. A climate-to-society pathway would have to run $\Delta T \rightarrow \Delta Y_{\text{crop}} \rightarrow p_{\text{food}} \rightarrow \text{affordability} \rightarrow \text{protest}$, and GIM now contains every link of that chain (Section 8), but the sweep plotted here enters at the second link so that the dose is controlled.

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